

Learning Management System integrated with an Applicant Pooling Module of JMH Microfinance Inc.

An Undergraduate Capstone Project
presented to the faculty of the
College of Information Technology Education,
Forbes College, Inc.

In partial fulfillment
of the requirements for the degree
Bachelor of Science in Information Technology

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Module of JMH Microfinance Inc.**

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Chapter 1

Introduction

Human Resources or HR is the division of a business that is charged with finding, screening, recruiting, training job applicants, and administering employee- benefit programs. An HR department is an essential component of any business, regardless of organization's size. HR departments are uniquely positioned to push strategic initiatives and drive employee engagement across the organization. As the custodian of talent, it is HR's responsibility to manage employee recruitment, on-boarding, development, and retention strategies to ensure that employees feel valued and engaged in their roles.

Technology has an important place in almost every business function. Whether you're communicating with your team, updating a policy or reviewing old records, technology has a major part in helping employees' complete jobs and progress in their careers. The technology that employees use includes computers, internet systems, printers, mobile devices and software applications that help them organize and prioritize work. Business technology can also help divide work as well, giving employees different goals to make larger tasks easier.[25] Technology is about innovation and innovation in business is all about doing things differently in order to provide better products and solutions, and an improved service to customers. Technology is not just essential for day-to-day businesses processes, but it can also help companies to achieve growth and success when utilized effectively. Technology can help businesses to achieve all of these objectives. Increasing productivity and efficiency is key to reducing costs and maximizing revenues, which can then be invested into growing the business. [12] Technology integration is the approach that companies use to choose and refine the technologies employed in a new product, process, or service. Technology integration has always been important, but in the past

ten years it has become much more important- and challenging- for obvious reasons. The number of technologies from which companies can choose has grown dramatically. Advances in chemistry, information technology, electronics, and materials science, for instance, mean that the technological bases of many industries are changing rapidly and unpredictably. [14]

A learning management system is a software application or web-based technology used to plan, implement and assess a specific learning process. It's used for e-learning practices and, in its most common form, consists of two elements: a server that performs the base functionality and a user interface (UI) that is operated by instructors, students and administrators. Typically, an LMS provides an instructor with a way to create and deliver content, monitor student participation, and assess student performance. It might also provide students with interactive features, such as threaded discussions, video conferencing and discussion forums. Businesses, government agencies, and traditional and online educational institutions often use these systems. They can improve traditional educational methods, while also saving organizations time and money. An effective system lets instructors and administrators efficiently manage elements such as user registration and access, content, calendars, communication, quizzes, certifications and notifications. The Advanced Distributed Learning group, sponsored by the U.S. Department of Defense, has created a set of specifications called the Sharable Content Object Reference Model (SCORM) to encourage the standardization of LMSes. [15]

Just like the LMS, the applicant pool is important in microfinance because it ensures that institutions can recruit qualified and capable individuals who can effectively support and advance their mission. A well-managed applicant pool helps microfinance organizations identify the best candidates for various roles, from loan officers to financial educators, ensuring they have the skills and knowledge necessary to deliver high-quality services. An applicant pool is a group of candidates who have applied for a particular role. It is also known as a hiring pool or a candidate pool. It's formed of anyone who has applied, from under qualified candidates to the most experienced ones who will likely get an interview. The employer's job is to take the hiring pool and narrow it down until the perfect candidate is chosen. The applicant pool varies in size depending on the type of job,

how much experience is required, and what the organization is. There are also variants when it comes to applies, too. If the position is a high-level senior role, the applicants will likely be experienced and highly trained. For jobs that require a little experience or qualifications, the hiring pool will be a little more mixed. [22]

1.1 Project Context

JMH Microfinance, Inc. is a mission-driven Non-governmental organization(NGO) operating in the Bicol Region,(JMHFI) was officially registered on October 7,2004, JMH Microfinance, Inc. is a regular member of the Microfinance Council of the Philippines, Inc., and compliant with the rules and regulations imposed by Microfinance NGO Regulatory Council (MNRC), Securities and Exchange Commission (SEC),They provide small loans, savings account, and insurance services to individuals who cannot access traditional banking services due to limited income, collateral, or formal credit history. With the goal of alleviating poverty and promoting socio economic development that aims to provide scholarships to deserving students and help alleviate poverty in the community.

JMH Microfinance uses various quizzes and evaluation methods to assess employee knowledge and skills, such as multiple-choice, scenario-based, and practical application tasks. These are complemented by pre- and post-training assessments to measure learning effectiveness, feedback surveys for continuous improvement, and performance-based evaluations to ensure employees can apply their training in real-world scenarios.

Refresher training needs are identified through performance evaluations, quiz results, managerial feedback, and incident reports, ensuring employees stay updated on policies and maintain high performance. To evaluate potential hires or low-performing staff, the Field Stress Test (FST) is used to assess their adaptability and resilience in handling high-pressure, real-life situations, preparing them for the demands of the job.

As the best solution to address the problem encounter, the proponent decide to develop a Learning Management System with an Applicant Pooling Module in JMH Micro-

finance, Inc. can address its current challenges in recruitment training, and employee reservation. This integration improves HR processes, enhance the effectiveness of training programs, lessen manual processes, and provides valuable insights for continuous improvement. LMS have different types of module

1.2 Purpose and Description

This study aims to addresses the organization's delay in recruitment process by streamlining the training process through an online learning system with a pooling module that will enhance efficiency, accessibility, and effectiveness in training practices, while also organizing and managing promising applicant data.

Benefits the following:

- **JMH Microfinance Inc.** - The system will help the company reduce operational disruptions and enhance productivity in response to the challenges caused by high attrition rates.
- **HR Department** - The system will help the department to reduce time spent and overall expenses associated with recruitment and training processes.
 - **Talent Acquisition Development Team** - The system will greatly help the team save time in finding the replacement for the vacant position and reduce the overall efforts and expenses used in the process.
 - **Training Development Officers** - The system will help the team identify and address the training needs. The system will also make the implementation and evaluation of the training.
 - **Trainers** - The system will help the trainers in tracking and evaluating the trainee's progress. The system would also allow them to easily add, hide and remove training modules.
- **JMH Trainees** - The system will allow trainees to access training modules from any location with internet connectivity, using their smartphones or personal computers.

They will also have the capability to track their progress during theoretical training sessions.

- **Researchers** - This will help the researchers gain practical experience in system design and implementation through the development of the Learning Management System, enhancing their expertise in technology-enabled solutions. Collaboration with JMH Microfinance Inc. will provide valuable insights into microfinance operations, facilitating the adaptation of similar systems to various contexts.
- **Future Researchers** - It will serve as a guide and reference for future researchers intending to undertake studies specifically focused on Learning Management Systems for Microfinance.

1.3 Objectives

1.3.1 Main Objective

The primary objective of this study is to develop a Learning Management System with a pooling module customized to the needs of JMH Microfinance Inc. This system aims to streamline training processes, enhance operational efficiency, and reduce both expenses and time spent on finding new employees.

1.3.2 Specific Objectives

- Develop an applicant pooling module that stores and ranks the grades, evaluations, and data of promising applicants, categorizing them into specific pools based on job roles and/or departments.
- Develop a progress tracking module that enables training officers to effectively monitor trainee progress, including attendance, login times, scores, and evaluations.

- Streamline the recording and management of training data and applicant pool data to ensure accurate, up-to-date records of trainee performance, progression, and applicant information.

1.4 Scope and Limitations

1.4.1 Scope

The study focuses on the development of an Online Product Monitoring and Inventory Management System for WeaveManila Inc., Sto. Domingo, Albay, concerning production status and inventory availability. The system includes the following features:

- **Dashboard Module** - This Module will be the main hub for users, showing a customized overview based on their roles. It will display key information like user activity, course progress, and system status, upcoming tasks, events and feedback. Simple visuals and alerts will help users easily find what they need and complete tasks.
- **User Management Module** - this will enable administrators to define user roles and permissions, ensuring secure access control within the system. It will facilitate account creation, authentication, and provide the functionality to promote or deactivate an accounts.
- **Login Module** - The Login Module in the LMS will serve as the gateway for users to access the platform. It will require authentication mechanisms such as username and password.
- **Automated Grading and Evaluation Module** - This module will manage the assessment process by automating grading and feedback for quizzes, tests, and assignments within the LMS.
- **Progress Tracking Module** - This module will allow learners to track their progress through courses and allow trainers to set objectives and milestones to guide learn-

ers. The module will include tools for tracking completion percentages, quizzes, and assignments. This module also allow trainers to monitor overall progress of the trainees courses and modules.

- **Applicant Pooling Module** - The applicant pooling module will centralize and streamline the management of job applicants within the system. It will provide administrators with tools to categorize applicants into specific pools based on job roles or departments.
- **Refresher Module** - This will offer current employees a way to refresh their knowledge and skills on topics relevant to their roles in the microfinance business. It will help them stay up-to-date and proficient, boosting productivity, job satisfaction, and overall performance.
- **Record Management Module** - This module will serve as a central hub for securely storing and managing various types of records. It will provide functionalities for creating, storing, retrieving, and organizing records efficiently.
- **Back-up and Restore Module** - The Backup and Restore module will maintain the continuity and reliability of educational content and user data. This will help administrators to schedule regular backups of course materials, user progress, and system settings to prevent data loss. In case of system outages or data corruption, the module allows for quick and efficient restoration of the LMS to its previous state, minimizing disruptions to teaching and learning activities.

1.4.2 Limitations

- **Essay automated assessment** - While the Learning Management System has the capability to automatically assess objective responses, such as multiple-choice or true/false questions, it lacks the functionality to evaluate open-ended, essay-style responses. This limitation arises because essay-type activities require subjective grading based on content, coherence, grammar, and argument quality, which cannot be accurately assessed by automated systems.

- **Practical Training** - Although the LMS effectively supports and assesses the theoretical aspects of training, it does not extend to the practical, hands-on components of training. This limitation means that while learners can acquire knowledge and complete assessments on theoretical topics, they must rely on separate, in-person training sessions or other practical platforms to gain hands-on experience and skill application.

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Chapter 2

Review of Related Systems and Related Literature

This chapter shows the related literature and studies of this research. Various Literature and studies conducted by other researchers have similarities and/ or opposing findings about this study which are presented in this chapter.

2.1 Related Literature

2.1.1 Technical and Didactic Knowledge of the Moodle LMS in Higher Education. Beyond Functional Use

1. According to Julio Cabero-Almenara, María Arancibia, Annachiara del Prete (2019) ,Higher education institutions at the international level have seen the need to adopt and integrate information and communication technologies to meet the opportunities and challenges of innovation in teaching and learning processes. This logic has led to the implementation of virtual learning environments called 'Learning Management Systems', the functionalities of which support flexible and active learning under a constructivist approach. This study measured didactic and technological use of Moodle and its implications in teaching from a quantitative approach by administering a questionnaire to a sample of 640 higher education teachers[5].

This system is related to the proposed system because thru virtual learning trainees use the modules to support interactive and collaborative learning experiences, functional use of the platform for the trainees that monitoring progress.

2.1.2 Future Trends in the Design Strategies and Technological Affordances of E-learning

According to Begoña Gros Francisco J. García-Peñalvo (2023) E-learning has become an increasingly important learning and teaching mode in recent decades and has been recognized as an efficient and effective learning method. The rapidly rising number of Internet users with smartphones and tablets around the world has supported the spread of e-learning, not only in higher education and vocational training but also in primary and secondary schools. E-learning and traditional distance education approaches share the emphasis on any time, any place learning and the assumption that students are at a distance from the instructor. The design of the initial e-learning courses tended to replicate existing distance education practice based on content delivery[13].

This system is related to the proposed system because it support the accessibility and convenience afforded by digital technologies in virtual distance learning.

2.1.3 Systematic Review of Automatic Question Generation for Educational Purposes

According to Ghader Kurdi, Jared Leo, Bijan Parsia, Uli Sattler and Salam Al-Emari(2020) while exam-style questions are a fundamental educational tool serving a variety of purposes, manual construction of questions is a complex process that requires training, experience, and resources. This, in turn, hinders and slows down the use of educational activities providing practice questions and new advances adaptive testing that require a large pool of questions. To reduce the expenses associated with manual construction of questions and to satisfy the need for a continuous supply of new questions, automatic

question generation techniques were introduced. This review extends a previous review on AQG literature that has been published up to late 2014. It includes 93 papers that were between 2015 and early 2019 and tackle the automatic generation of questions for educational purposes[16].

This system is related to the proposed system in terms of educational tool, to train the newly hired by posting exams, quizzes and other training modules to monitor the trainees.

2.1.4 Effect of learning management system on Student's performance in educational measurement and evaluation

According to Basil C. E. Oguguo, Fadip Audu Nannim, John J. Agah, Christian S. Ugwuanyi, Catherine U. Ene Augustina C. Nzeadibe (2020) the effect of Learning Management system (LMS) on students performance in educational measurement and evaluation course. A non-equivalent group quasi experimental research design was adopted for the study, using a population of all the undergraduate students in Imo State University, Nigeria. A sample of 232 students was purposively selected comprising of 109 males and 123 females. The researchers recommend that, LMS Packages such as Moodle should be learnt and used by lecturers of Educational Measurement and Evaluation[2].

This system is related to the proposed system in terms of evaluating the newly hired, for the current manual system was a time consuming process and was not reliable anymore. By the proposed system the JMH Microfinance will going to save a lot of time.

2.1.5 Learning Management System Use with Online Instruction

According to Bradley, Vaughn Malcolm (2021) the learning process through online classroom environments. A standard LMS supports an inclusive learning environment for academic progress with interceding structures that promote online collaborative-groupings, professional training, discussions, and communication among other LMS users. Instructors should balance active learning with the use of LMS technological resources and the

use of guidelines from the qualified curriculum. An LMS allows instructors to facilitate and model discussions, plan online activities, set learning expectations, provide learners with options, and assist in problem-solving with processes for decision making[8].

This system is relate to the proposed system because this system has a database which trainees can view their learning materials and also the HR can store the modules.

2.2 Related System

A systematic review on trends in using Moodle for teaching and learning

The Moodle Learning Management System (LMS) is widely used in online teaching and learning, especially in STEM education. However, educational research on using Moodle is scattered throughout the literature. Therefore, this review aims to summarise this research to assist three sets of stakeholders—educators, researchers, and software developers. It identifies: (a) how and where Moodle has been adopted; (b) what the concerns, trends, and gaps are to lead future research and software development; and (c) innovative and effective methods for improving online teaching and learning. The review used the 4-step PRISMA-P process to identify 155 suitable journal articles

2.3 Related System

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Moodle has been widely adopted for online learning in diverse educational settings, which showcases its flexibility and scalability. This adaptability can be relate in our study, where integrating an LMS with a recruitment module needs a system capable of handling both educational and corporate functionalities. The scalability of Moodle could serve as a reference for integrating recruitment tools within an LMS designed for various organizational needs, such as those of JMH Microfinance.

2.3.2 A review of learning management systems (LMS) framework towards the element of outcome-based education (OBE) (Haziman Madiyah, Rosmayati Mohemad - March 2023)

The emergence of the Learning Management System (LMS) has significantly enhanced the teaching and learning process by centralising the management, organisation, delivery, and reporting of educational content and student outcomes. Moving towards outcome-based education (OBE), a LMS that is capable of supporting the delivery of OBE courses can facilitate more effective engagement with students, thereby improving students' learning outcomes and accelerating the process of continuous quality improvement. However, there is no well-rounded LMS framework for supporting the implementation of the OBE learning process to the best of our knowledge[20].

The study on integrating Outcome-Based Education (OBE) into Learning Management

Systems (LMS) shows the need for an efficient framework to manage courses, user interaction, and assessment. For our project, can take away key strategies such as creating a user-friendly interface and continuous improvement tools, which can also apply to integrating a recruitment module into the LMS for JMH Microfinance. The insights into improving engagement, usability, and supporting different user behaviors will help ensure the Applicant Pooling Module is effective and easy to use for HR staff and applicants.

2.3.3 Advanced Applicant Tracking Systems (ATS) (Prasad R. Chavan, Yash Chandurkar, Rohit Chavan - July 2024)

The Applicant Tracking System (ATS), also known as a talent management system or job applicant tracking system, is a software application designed to facilitate more efficient recruitment processes for companies or selection agencies. The objective of ATS is to streamline various aspects of the recruiting process, from receiving applications to hiring employees and effectively manage recruitment needs electronically. Methodologies such as NLP and KNN models are used for automated resume parsing and classifying the resume from unstructured format to structured format. The final results found significant improvement in performance of functionalities such as candidate screening, applicant testing, interview scheduling, managing the hiring process, reference checks, and completing new hire paperwork[9].

The study on Advanced Applicant Tracking Systems (ATS) shows how automating tasks like resume parsing, candidate screening, and interview scheduling can make recruitment faster and easier. For our project, integrating these ATS features into the LMS for JMH Microfinance can streamline the recruitment process, saving time for HR staff and making hiring more efficient.

2.3.4 Digital Job Searching and Recruitment Platforms: A Semi-systematic Literature Review (Signore, C., Della Piana, B., Di Vincenzo, F. (2023))

This study is to shed light on the new E-recruitment trend that is pervading the lives of job seekers, included students, and job offers. A semi-systematic literature review on digital job searching and recruiting platform in the last five years was conducted with the aim to develop a preliminary conceptual framework. Following a replicable research process, a final sample of 37 publications was located in five subdimensions - Web Application Framework, Use of Artificial Intelligence technologies, Use of Blockchain Technologies, Type of User, User Experience - grouped by two dimensions of analysis: “Technical implementation of the platform”, “Platform usability analysis”[23].

The study on Digital Job Searching and Recruitment Platforms looks at the growing trend of e-recruitment and its impact on job seekers and employers. For our project, these insights can help guide the integration of modern tools like online resume screening and blockchain for secure data handling, improving both the efficiency and usability of the LMS and Applicant Pooling Module for JMH Microfinance.

Chapter 3

Technical Background

This chapter represents the technical background of the study and discusses the project methodology used in constructing and developing the proposed study. It includes the project development, resources, and research made for the proposed system.

3.1 System Architecture

The Learning Management System (LMS) is designed to provide an online platform for employees, Applicants, and administrators to interact, manage, track progress, and access job materials. The system is accessible through web browsers and devices, ensuring ease of access and usability.

The platform also hosts the LMS's MySQL database, which stores user accounts, applicant data, scores, and activity logs.

3.2 Software and Technologies

The proponent used the hereunder technologies for the development of the system:

- **Visual Studio Code 1.83.1:**

Visual Studio Code, also commonly referred to as VS Code, is a source-code editor developed by Microsoft for Windows, Linux and macOS. Features include sup-

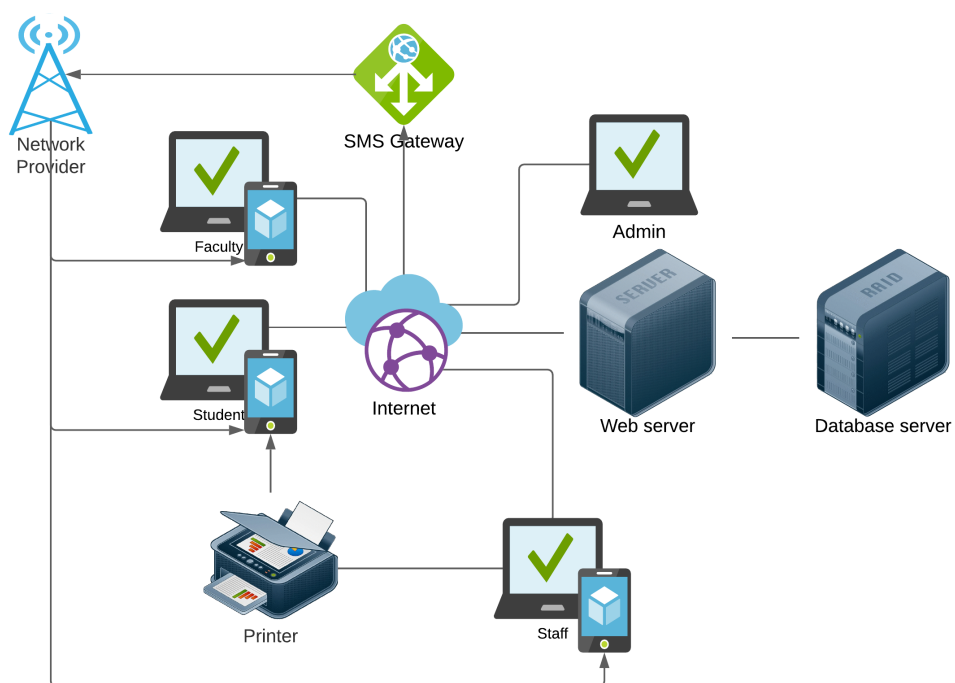


Figure 3.1: The System Architecture of the Proposed System

port for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git. Users can change the theme, keyboard shortcuts, preferences, and install extensions that add functionality.[21]

We will use the Visual Studio Code software for constructing the system.

- **HTML5.0:**

HTML5 (Hypertext Markup Language 5) is a markup language used for structuring and presenting hypertext documents on the World Wide Web.[7]

The HTML will be used to create the structure and content of the web pages.

- **CSS Version 3**

Cascading Style Sheets (CSS) is a style sheet language used for specifying the presentation and styling of a document written in a markup language such as HTML or XML. CSS is designed to enable the separation of content and presentation, including layout, colors, and fonts. [18]

The CSS will be used to control the visual styling of the web pages.

- **JavaScript ES14:**

JavaScript often abbreviated as "JS", is a programming language and core technology of the World Wide Web, alongside HTML and CSS.[10]

By using the JavaScript, it will add a interaction for the web pages.

- **PHP 8:**

Hypertext Preprocessor (PHP) is a general-purpose scripting language geared towards web development. On a web server, the result of the interpreted and executed PHP code which may be any type of data, such as generated HTML or binary image data would form the whole or part of an HTTP response.[17]

By using PHP the team can interact with MySQL database to retrieve and manipulate data and generates HTML, CSS, and Javascript code to display the system or website

- **MySQL Database:**

MySQL is an open-source relational database management system (RDBMS). It enables user to store, manage, and retrieve structure data efficiently.[4]

The MySQL Database is a common and easy-to-use data storage for web pages, therefore we decided to use it as a database for our system.

- **ReactJS 18:**

ReactJS, also known as React, is a popular JavaScript library for building user interfaces. It is also referred to as a front-end JavaScript library. It was developed by Facebook and is widely used for creating dynamic and interactive web applications. In this article, we'll explore the key concepts of React.[11].

By using the ReactJS as framework, it will help the developers to create maintainable and fast user interfaces.

- **XAMPP 7.4.1 :**

XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages.[1]

The XAMPP are common and famous local host or server, and we decided to use it as a server for our system

3.3 Hardware and Software Specifications

The proponent used the hereunder software and hardware specifications in developing the system:

Hardware

- Processor 1.6 GHz dual-core processor (minimum)
- 2.4 GHz quad-core processor (recommended)

- 512GB Solid-State Drive
- Memory (RAM) At least 2 GB (recommended: 4 GB)
- PC/Laptop

Software

- Operating System Windows 10 (Version 1903 or later)
- Web Browser: Google Chrome (Version 83 or later)

3.4 Technical Features

The system will first be developed and implemented using the proponent's computer, which when finalized, will be housed in a dedicated web server. The SMS notification will be catered by an online SMS gateway. The following are the specific technical features of the system:

- **SECURITY:** Users need to log in to their account for proper authentication prior to utilization of the system. For security measures, the passwords stored in the database are encrypted using PHP's Bcrypt hashing. To ensure the security of the website, the server is implemented with SSL (Secure Socket Layers) which is responsible for the encryption of data sent between the server and the client computer.
- **Forgot Password Module**
- **AUTO-SAVE:**
- **GENERATION OF STUDENT SCHOLASTIC DOCUMENTS:**
- **SMS NOTIFICATION**
- **DATABASE BACKUP AND RESTORE MODULE:** The system has a backup and restore module in case of an unexpected event. If in case corruption of the database

occurs, the system is able to get back with the restore module. The admin only needs to choose which SQL file to use when restoring the database.

- **Logs and Audit Trails Module**
- **FEATURE D:** Explanation as to how it functions
- **FEATURE E:** Explanation as to how it functions

Chapter 4

Methodology

The proponent used the RAD (Rapid Application Development) Methodology to develop the system. The Methodology reduced the planning time and had the client check the prototype as well as the irregularities of the system. It allowed the client to measure the progress, and to communicate with the developer from time to time. This methodology helped to lessen the complexity of the development of the entire process [3].

Figure 4.1 shows the RAD Methodology. This is how the development cycle of RAD works.

[]

4.1 Requirements Planning

In this stage, the proponent gathered data from the client through an interview. The process included identifying the current processes, and problems encountered in the institution and analyzing what solution can be made out from the problems elicited in terms of their grade submission and approval, efficient production of documents requested by students, and record-keeping of grades of students. The current process of the client is illustrated in Figures 4.2 to 4.5. Upon investigating the current state of the matter, the scope and limitations for the project as well as current and potential issues that would need to be addressed during the development is also determined.

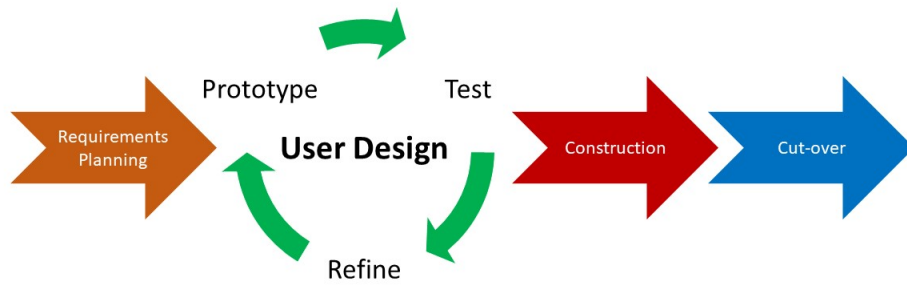


Figure 4.1: RAD Model

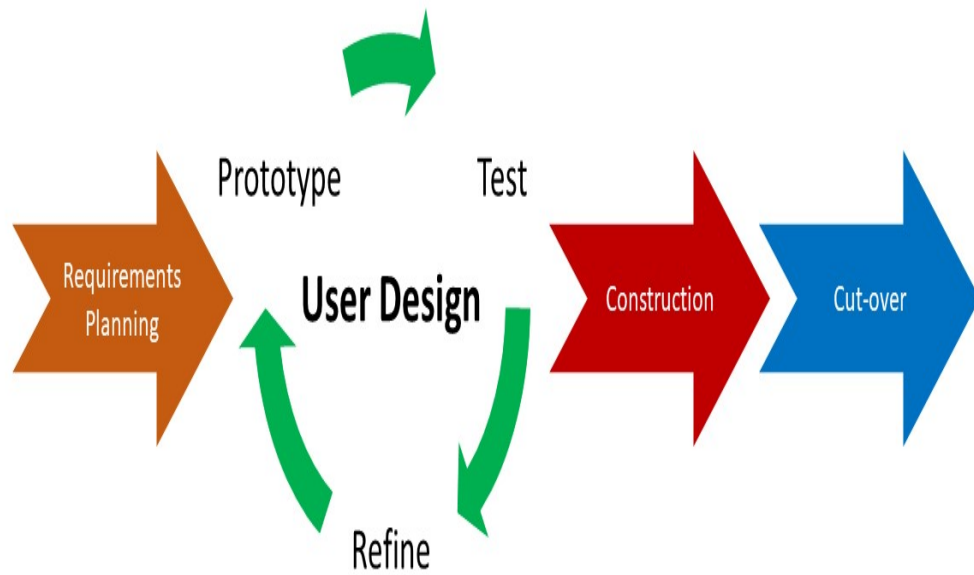


Figure 4.2 depicts how the current process of enrollment at School XYZ works. As most of the processes in the stated college are manual, the process starts with students filling up their information in a form. The registrar will then check the form to verify the completeness of the student details as well as the documents for enrollment. Only students with completed details and requirements get to proceed to the cashier, where a statement of account will be given to them. Students can pay partially or in full, and they will be given a receipt to be presented to the college registrar before they can be truly enrolled.

While the process of enrollment is still paper-based, the college registrar stated that the institution has not experienced major problems with its current process.

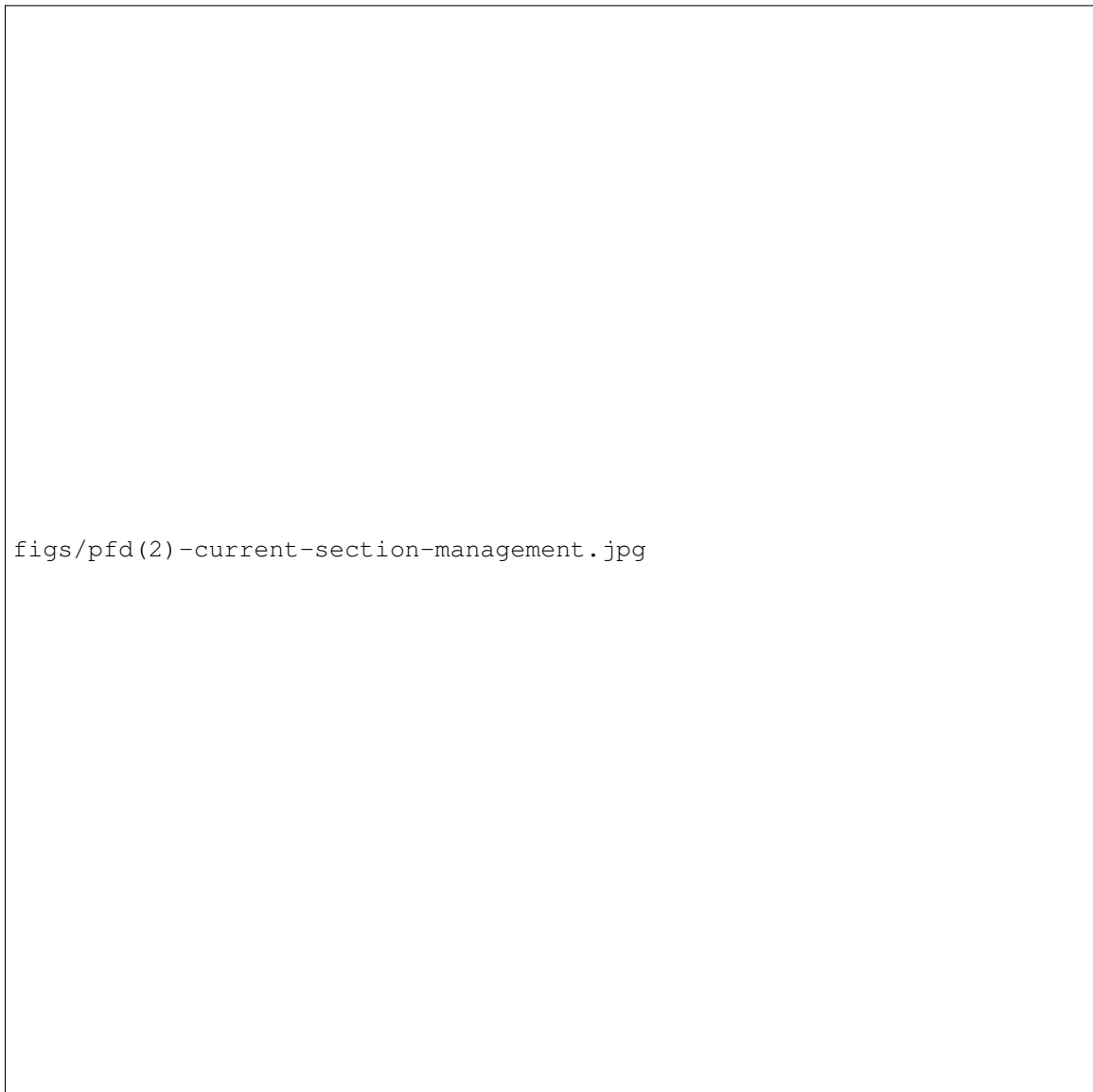


Figure 4.3: Activity Diagram: Section Management

Figure 4.3 shows the current process of how sections are managed in the institution. This process takes place and can effectively commence after the enrollment period. In the above figure, the registrar is responsible for distributing the courses and sections handled by instructors. It is then the responsibility of the instructors to enlist all students enrolled

in each handled section. To do so, students have to present their enrollment form, duly signed by the registrar.

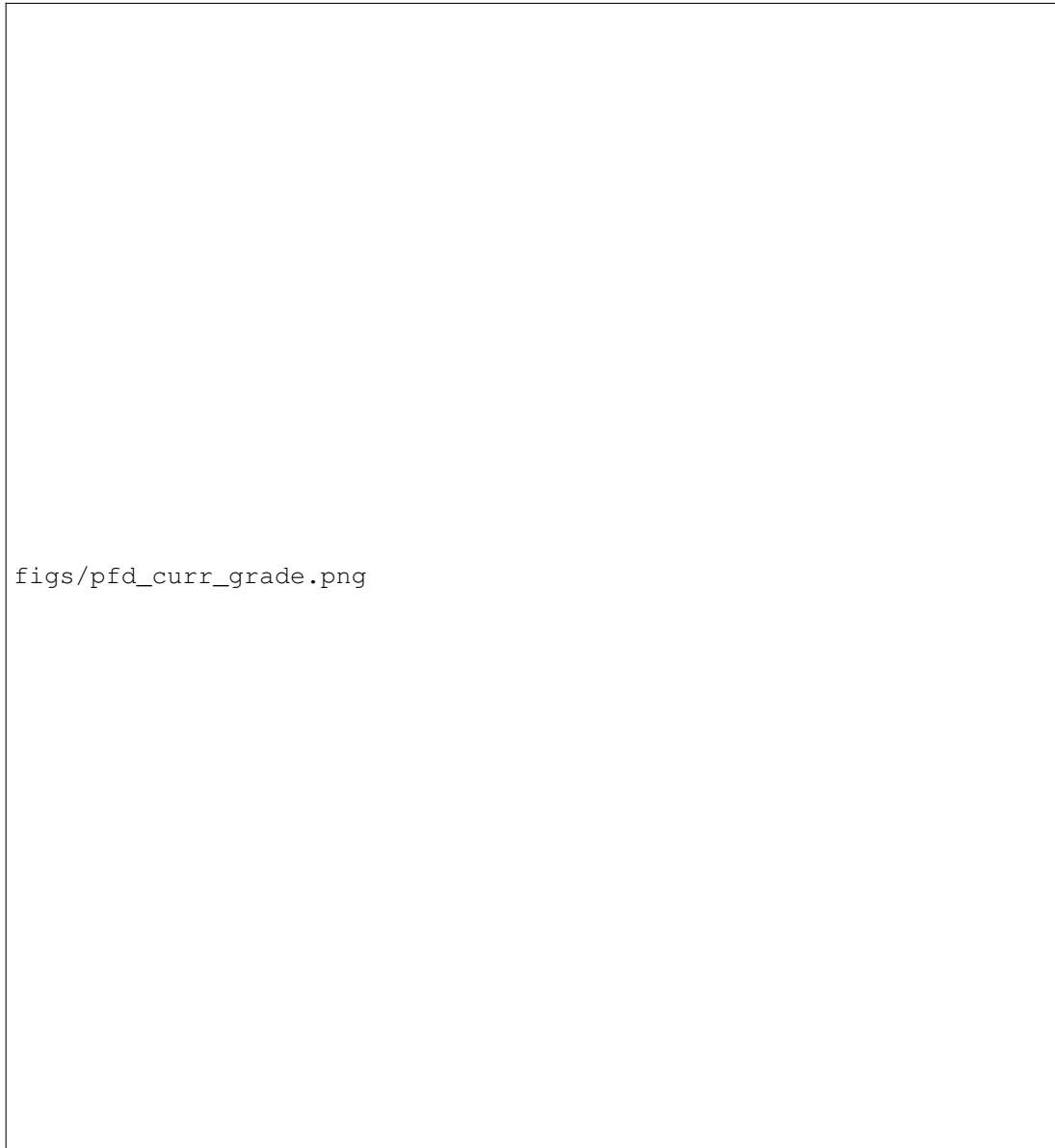


Figure 4.4: Activity Diagram: Student Grade Submission and Approval

Figure 4.4 explains how the current process of grade submission works. The process

starts when instructors compute and print the grades of the students. The printed grades will then be forwarded to the registrar, where the contents will be checked for veracity. Should there be discrepancies or errors in the document, it will be returned to the instructors for resubmission. Should the grade record be accepted, the record will then be forwarded to the registrar's staff for document reproduction for a copy of the registrar's office and the record custodian. The submitted grades will then be encoded by the staff into an Excel file.

According to some instructors, there were cases when students complained about not having submitted their grades to the registrar's office when, in fact, they had already done so. The registrar's office took time to generate a student's scholastic request due to the lack of grade records for some subjects taken by students.

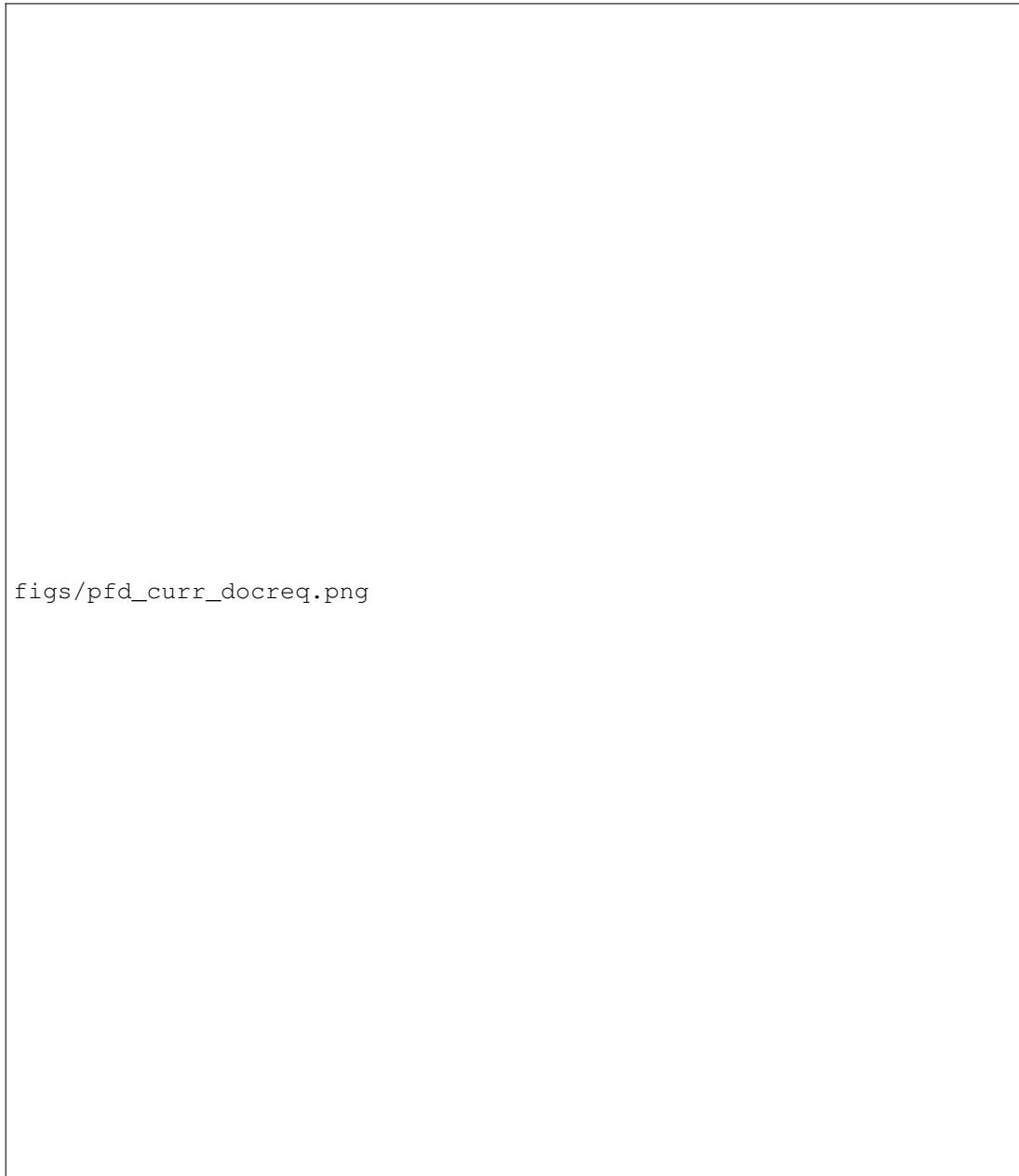


Figure 4.5: Activity Diagram: Student Document Request

Requests related to scholastic documents start with students inquiring with the Registrar's Office. The office is then responsible for giving out a form that should be filled

out by students. Students will then be asked to proceed to the cashier's area, where the student's form will be checked and payment will be settled. After getting the receipt, the student will then return to the registrar's area, where a claim stub will be given. The requesting students will then be informed to return after a week for the release of the requested documents.

4.2 User Design

In this phase, the proponent presented a prototype of the system and discussed the functionalities with the registrar and administrator. This step was conducive to both the proponent and the main users in order to have a starting point for the development of the system. After the deliberation, the proponent and registrar agreed to the modules that will be developed for the system. Hereunder are the agreed-upon main modules to be developed in the system:

- Streamlining of student grade submission and approval
- Management of student grades
- Generation of scholastic documents particularly the Certification of Grades and Transcript of Records

Prior to the start of the prototype, the proponent prepared a flowchart that illustrates how the new system operates. The proposed operation of the new system is depicted in Figures 4.6 to 4.8

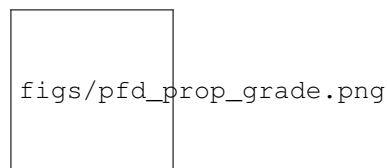


Figure 4.6: Activity Diagram: Proposed Grade Submission and Approval

Aside from its centralization of student grade records, one of the purposes of the proposed system is to simplify the submission and approval of student grades.

The process commences with instructors retrieving all respective sections handled. For each term, instructors will be tasked with inputting the students' grade records. The system includes an auto-save feature, which is beneficial in case a sudden power interruption or intermittent connection takes place. This will eliminate the need for the instructor to re-input the grades in the section being worked on. After inputting all student grades, the instructor shall then proceed with submitting them to the registrar. Once submitted, the instructor cannot edit the term grade. The system will inform the registrar of a grade submission via system and SMS notification. Should the registrar reject the submission, the instructor can re-input student grades and repeat the submission process; otherwise, the term grade is finalized.



Figure 4.7: Activity Diagram: Proposed Student Document Request

In the developed system, as shown in Figure 4.7, requestors will have to fill out a form that will ask for the student's record and the requested documents. To provide additional authentication measures, the requestor's mother's maiden name and the student's date of birth will also be needed in the form. If the entry does not match the student's

record, the request will not commence. Only requests with veracious entries are accepted by the system. In this way, the system can prevent unwarranted spam requests. Upon acceptance, the system will proceed to send an SMS to the system indicating the expected amount for payment and the next step, which is to pay the cashier. On the cashier's part, the system will display all requests that are pending for payment. Once the student has paid the required amount, the cashier needs to update the status of the document request with the click of a button. The requestor shall receive an SMS code that one can use to monitor the status of requested documents. Since one of the problems students face regarding scholastic requests is the need for constant follow-ups to know the status of their documents, this feature will be valuable to conserve time and money by eliminating the need to physically go to the institution. After updating the request from the cashier's side, the requested document shall be in the care of the registrar's office for generation and release.



Figure 4.8: Activity Diagram: Proposed Document Generation and Tracking

The process of document generation starts with the system retrieving and displaying all pending document requests for processing. Only paid requests are visible for the generation of documents by the registrar's office. For scholastic documents such as certificates of grades and transcripts of records, the system will retrieve all student grade records in the system. This will tremendously reduce the amount of time needed to generate a document since grade records are centrally stored in the system. Once printed, the said office is responsible for creating a tracking history of the document. This is needed in order for the requestor and related users alike to monitor the location and status of documents in particular. The generated document will be forwarded to the administrative office for signature and then returned to the registrar's office for release. In the process of forwarding the documents, the forwarding personnel needs to record the details, particularly the receiver and datetime of their receipt. Once all requested documents are processed, the system will inform the user via SMS. In the Registrar's Office, all pending documents for release will be displayed.

More diagrams were created to aid in analyzing the proposed system. These include the Context Flow Diagram (*See Figure 4.9*) and Data Flow Diagrams (*See Figure 4.11 - 4.21*) which was used to understand who inputs, processes, and accesses data. The proponent also utilized an Entity Relation Diagram in order to systematically analyze the data requirements (*See Figure 4.22*).

figs/context-flow-diagram.jpg

Figure 4.9: The Context Level Data Flow Diagram

Figure 4.9 illustrates a high-level view of a system. This shows how data is inputted, saved, and accessed in the system. Specific details on the involvement of each actor in the system are as follows:

- Admin IT: The admin IT has the initial account in the system. A default account will be used to gain access to the system. To create other accounts that will be responsible for other functionalities, the admin IT will have to input the personal information and designated username of another principal account, such as the HR Staff. Having the privilege to manage accounts, admin IT has the luxury of viewing all existing accounts in the system.

Further, Admin IT is responsible for ensuring the data backup of the system. When providing a backup, the system prepares a downloadable SQL file that will also be used for restoring the system's content based on the latest backup.

- HR Staff: As with Admin IT, the HR Staff is responsible for the management of user accounts and personnel information.
- Registrar: The registrar's account is involved in most of the system's processes. As shown in ??, the registrar is responsible for managing student information, subject details, school details, student grades, section details, course details, and academic year details. As such, data for the involved functionalities comes from and is retrieved by the registrar. Other details are also passed to and from the registrar, which includes requested document details, history of document tracking, documents on hold, documents for release, and submission status.
- Registrar Staff: As personnel under the supervision of the registrar, the registrar staff is involved in some of the functionalities performed by the registrar, namely: management of student information, subject details, school details, management of student grades, section details, and document request management. As such, the registrar staff accesses all relevant information needed in the stated modules. However, in relation to grade management, the registrar staff does not have the authority to approve the grades submitted by instructors. As with the registrar, registrar staff are also responsible for the release of the documents.

- **Instructor:** The role of the instructor is to submit the student's performance in the form of grade records. Prior to the submission, instructors will have to access their corresponding sections. For each section, instructors must enlist all students enrolled in the system. Submissions are based on term periods, so each section requires four (4) submissions to properly give grades to the students. Each of these submissions will require the approval of the registrar. Further instructors can calibrate the term weight if the need arises.
- **Cashier:** Cashiers are responsible for the management of the details of documents requested by students. Intrinsically, the cashier can access and modify all relevant details, including the document name and its associated costs. Further, student document requests are handled initially by the cashier. If the request is paid, the cashier needs to update the status of the request before such a request is forwarded to the registrar's office. Cashiers can also directly request documents in the system on behalf of the student.
- **Admin Staff:** The admin staff has the least involvement in the system. They are mostly responsible for knowing what documents are currently in their possession. The documents need to be signed by the proper personnel before they are forwarded back to the registrar's office.
- **Student:** Students are involved with the request for scholastic documents. To initiate a request, a student will be given a form to fill out, including the type of document to request and its associated costs. If the information entered is accurate, the system will send an SMS to the student stating the amount to be paid to the cashier. Once paid, another SMS will be received containing the tracking code for monitoring the status of the requested documents.

figs/dfd-level-1.png

Figure 4.10: The Level 1 Data Flow Diagram

4.10 depicts a general overview but goes with more detail than the Context Level Flow Diagram. In this figure, the single process shown in Figure 4.9 is broken down into the main modules of the system. The details of these modules are as follows:

1. Account and Personnel Management: This module is responsible for searching, modifying, and creating the personnel and accounts of each involved personnel in the system. Each account holder can change their respective passwords. Further, this module is also responsible for the authentication and authorization of users who log in to the system.
2. School Management: The school management module is utilized to register the different schools that the system will use to record the grade records of transfer students.
3. Course Management: The course management module is implemented for managing the courses, in this case, the programs offered by San Jose Community College. In practice, all students are enrolled in a single program offering (course) in SJCC, thus requiring the student records to be associated with a course. In addition, this module can also update which courses will be discontinued.
4. Subject Management: In relation to the course, each respective course offers comprehensive subjects required to be taken by students for its completion. In the system, subject details are necessary to create sections where students are enrolled and handled by instructors.
5. Student Management: The proposed system is implemented with a student management module to effectively maintain student records. Aside from the usual record-keeping, the student's data, and the mother's maiden name will be used for authentication should students request for scholastic documents.
6. Academic Term Management: SJCC follows a regular semester academic calendar. The proposed system integrated an academic term module for adding and updating academic terms. The system manages an active academic term that will be followed when creating the sections for student enrolment.

7. **Section Management:** The section management module is responsible for creating and updating sections through which students are enrolled. Each section is handled by an instructor, associated with a subject, and is offered in a particular academic term. In the system, the active academic term serves as the default school year and term by which a section is created.

8. **Grades Management:** In this module, the instructors, registrar, and registrar staff are involved. This module includes a grade submission by instructors which can be done every grade term (i.e., Prelim, Midterm, Pre-final, and Final). Submissions made are approved/disapproved by the registrar. Instructors have different preferences on the weight of the term grade terms thus the system has a module for editing the term weight (percentage).

The registrar and registrar staff can encode the student grades into the system. This is done when recording the subjects taken and grades earned by a transfer student from the previous school.

9. **Document Management:** The document management module is utilized to update the prices of documents requested by students.

10. **Document Request:** This module is usually initiated by a student requesting a scholastic document. If the request is done by a student, the inputted values in the form will be validated against the data of the student in the database. Only requests with accurate entries will be recorded in the system. Upon successful request, the student will receive an SMS notifying of the next process and the amount to be paid to the cashier.

11. **Document Generation & Tracking:** The document generation and tracking module can commence once the cashier approves the document request from a student. Most of the process involved in this module is done by the registrar and/or registrar's staff. The content of the Certification of Grades and Transcript of Records can be generated based on the records from the grades management. All scholastic documents generated must be registered for document tracking before forwarding to other personnel or units involved (usually for signing). For each request, the system

will track the status per document and shall notify the requestor if all requested documents are available for release.

An effective information system cannot exist without data stores, responsible for the persistent storage of records. In the above figure, these data stores logically represent the storage of data received and accessed by all modules in the system.

Further discussion of the above modules will be provided below.

figs/dfd(1)-account-and-personnel-mgmt.jpg

Figure 4.11: DFD Module: Account and Personnel Management

The management of account and personnel information starts with the admin IT/HR staff registering the personnel information and temporary user credentials for the users. To proceed with the account creation, personnel details such as staff number, full name, unit or department assigned, position or rank, contact number, and username are required. The availability of staff numbers and usernames will be checked. If any of the usernames or staff numbers are already in use, the registration shall not proceed. The temporary password given to users is computer-generated and will not be visible to the person registering the account. The temporary user credentials will then be sent to the intended recipients via SMS. All users have the capability to change their passwords and update their personal information.

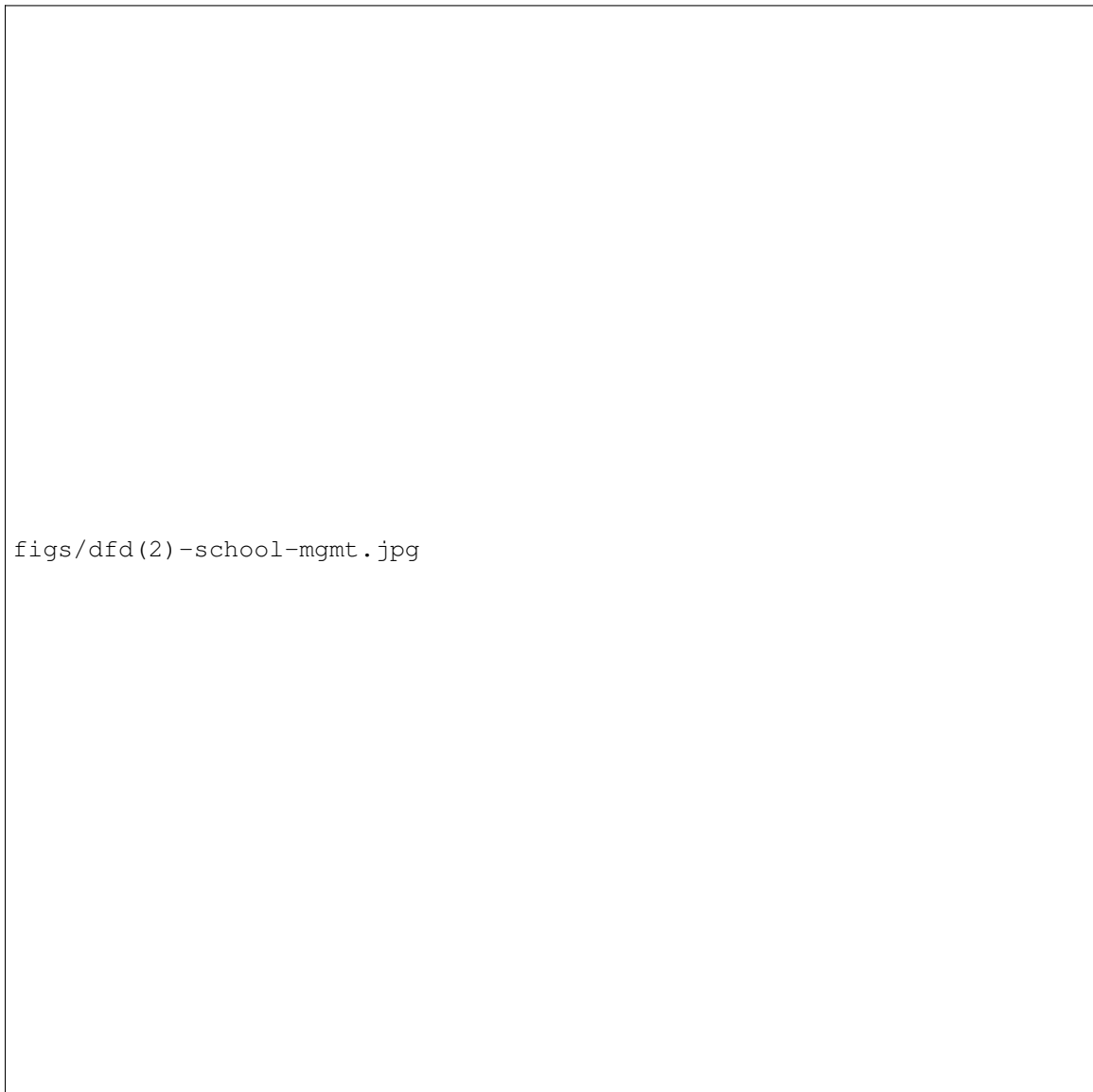


Figure 4.12: DFD Module: School Management

Figure 4.12 shows that both the registrar and the registrar staff have the authority to manage the school details in the system. The school details are important in such a manner that they are used in entering the credentials of students hailing from other institutions.

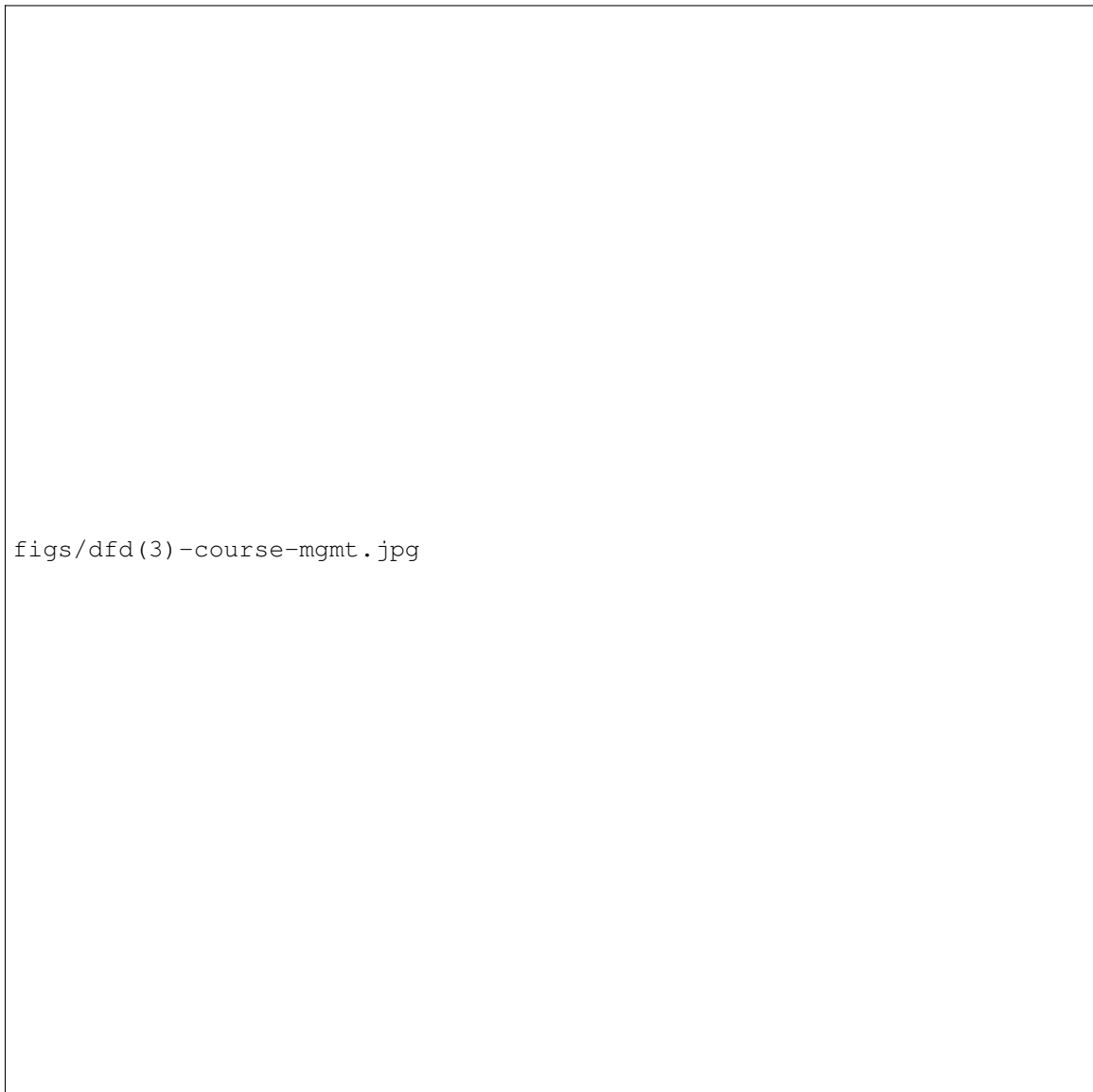


Figure 4.13: DFD Module: Course Management

Figure 4.13 depicts the data flow of how courses are managed in the system. Only the registrar has the sole authority to manage the courses in the system. The system starts by displaying all the courses available. For the registration of courses in the system, the registrar needs to input the course code, description, and major (if applicable),

as well as the department ID, which corresponds to the department that handles the inputted course. When updating the course details, the system will fetch the course ID (its identifier) and update its details with the corresponding data entered in the database.

figs/dfd(4)-subject-mgmt.jpg

Figure 4.14: DFD Module: Subject Management

In this module, both the registrar and registrar staff have the authority to manage the subjects. All subjects and their corresponding details will be listed on the interface. For the addition of new subjects, details, particularly the subject code, description, unit and lecture units, and special subject identifier, will be required. The system is responsible for ensuring that all subjects have a unique subject code; thus, registration of new subjects will not proceed if the subject code entered already exists. In cases where subjects are no longer offered, they can be deactivated.



Figure 4.15: DFD Module: Student Management

The registrar's office is responsible for managing student information. The student information required is: student number, student's full name, birthdate, contact number, course (retrieved from the course data store), and mother's full maiden name. The student's birthdate and mother's maiden name are crucial, as they will be used as additional

authentication measures for requesting student documents online.

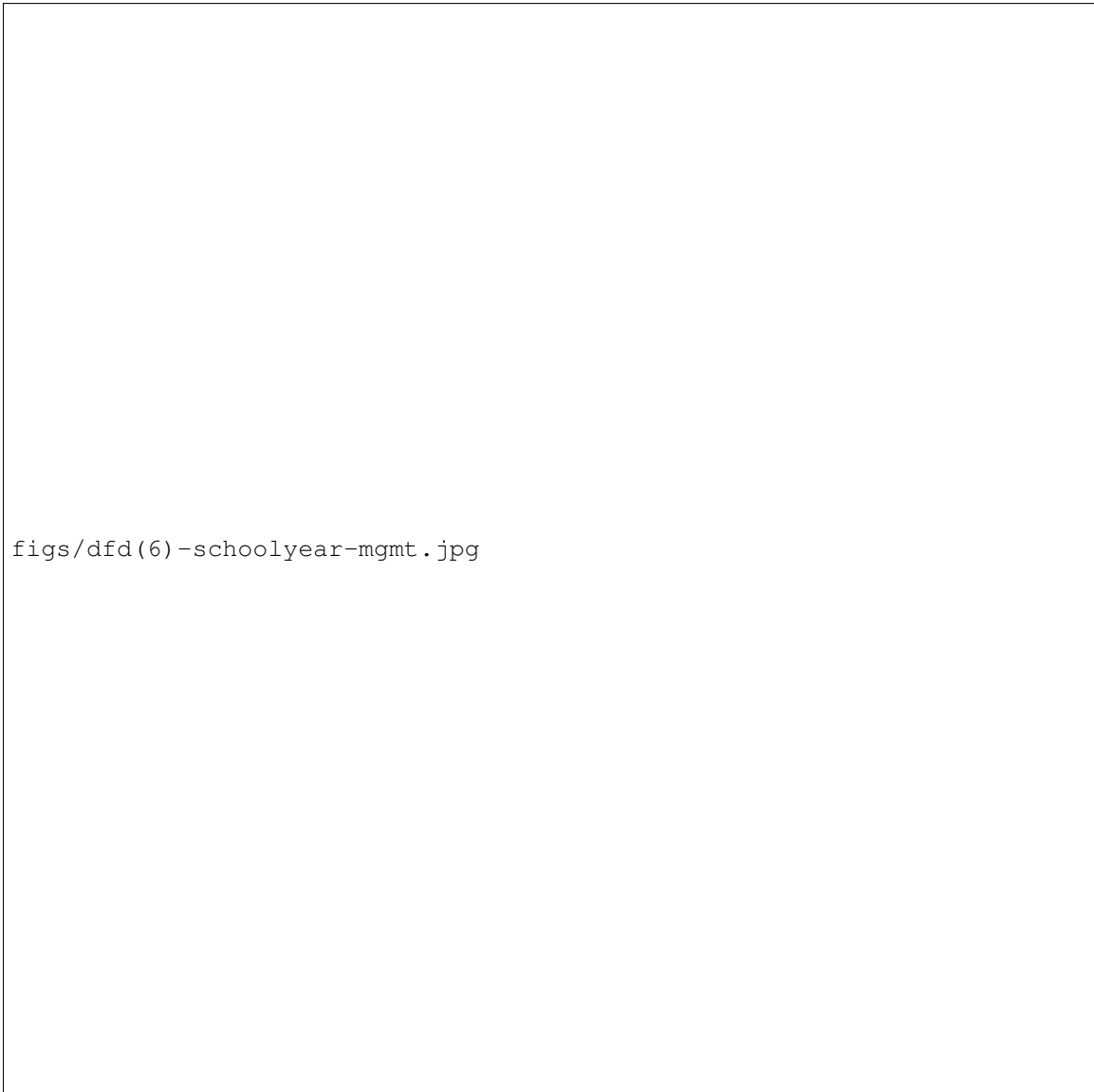


Figure 4.16: DFD Module: Academic Term Management

Only the registrar has the sole authority to manage the academic term. This module is used as a basis for the creation of sections. When accessing the section module, the default school year and term are based on the active school year. The first and last day of

classes shall serve as a reminder for the registrar of the need to shift to another academic term.

figs/dfd(7)-section-mgmt.jpg

Figure 4.17: DFD Module: Section Management

In this module, both the registrar and the registrar staff can manage the sections. In order to register and update a section, the system requires: (1) the subject to be handled; (2) the instructor handling the subject; (3) the course block that will be handled; and (4) the school year and term. Thus, there is a need to fetch stored data from the respective data stores. The system will be responsible for auto-generating a section name based on the user inputs.

figs/dfd(8)-grade-mgmt.jpg

Figure 4.18: DFD Module: Grade Management

The management of student grades involves the instructors, the registrar, and the registrar staff. All the handled sections of instructors will be listed in the dashboard. For all of the sections handled, instructors need to enlist all the students enrolled. The system ensures that no duplication of student numbers gets recorded in the same section. Prior to the submission of grades, an instructor has the freedom to set the term weight (e.g., 15% prelim, 20% midterm, 25% prefinal, 40% final). The system shall not accept a term weight that does not equal 100 points. While inputting student term grades, the system automatically saves (intervally) the current term grade being worked on.

In terms of the grade submission, the term submitted and the particular section are recorded in the data store. Once submitted, the instructor cannot edit student grades for that term. This section, term, and status will be retrieved for the registrar to view which section and what term have been submitted. As stated in refuse-case-grade, the registrar has the sole authorization to check the instructors' submissions. Whether approved or rejected, the system will update the submission status of the section and term submitted. If the grade submission is rejected, the instructor is notified and can edit and re-submit the student grades. Otherwise, the grades submitted are finalized. However, an instructor can appeal for the updating of student grades, but it will be the registrar who can input the updated grades.

The registrar staff can encode student grades that have been recorded prior to the implementation of the system. This will require the subjects taken and grades incurred by students.



Figure 4.19: DFD Module: Document Management

The management of documents to be requested is solely assigned to the cashier. In order to add a new document for request, the name of the document and its cost per page is needed. The same document details are required for updating the document item. If in any case, a particular document is not anymore offered, the cashier can remove the

document in the available document request list.



Figure 4.20: DFD Module: Document Request

The process of a document request is usually initiated by a student or requestor filling out the information on the web system. The form requires student information, including the student's number, the student's full name, course, student's date of birth,

and the full name of the mother's maiden name. If all inputted data matches a student record in the database, the system will accept such a request. The system will then compute the expected amount to be paid by the requestor based on the document request details before recording the document request details in the data store. The said amount will be sent to the requestor via SMS, including the next step to be performed. All pending payment document requests are listed in the cashier's dashboard. Once the student has paid for the request, the cashier will approve it, which will then be forwarded to the registrar's dashboard. An SMS containing the tracking code will be sent to the requestor.

figs/dfd(11)-document-tracking.jpg

Figure 4.21: DFD Module: Document Generation and Tracking

In this module, both the registrar and registrar staff can initiate the generation of documents. All pending documents for generation will be listed on the dashboards of both stated users. Once the documents are generated, the registrar or its staff must register the document for tracking. This is required so that the involved users in tracking can monitor the location and status of the documents. To proceed with the processing, the documents may be forwarded to the corresponding unit, which must sign them. After receiving, the forwarding personnel is responsible for updating the tracker by inputting the recipient and the datetime when the document was received. The process of forwarding is repeated until all the signatories required for the document have signed. At any given time, the requestor may use the tracking code to monitor the current holder of all requested documents. Once all documents involved in the request are completed, the system will send a notification to the requestor regarding its availability and release. Once released, the releasing personnel shall update the status of the document release.

figs/updated-erd.jpg

Figure 4.22: The Entity Relationship Diagram of the Proposed System

The *department table* is responsible for recording the involved departments where teaching personnel are assigned. Each of these departments is headed by a single person. The department houses one or more courses that are taken by students. *Courses* include unique course codes and descriptions. In the system, course major may be empty as not all courses have a major or field of specialization. On the other side, the unit table contains records of areas where non-teaching personnel are assigned.

The *school table* contains the school name of the school itself and other schools where transfer students hail from. The table is necessary to determine in what school a student took a grade record.

The *subject table* contains records of disciplines taken by students. This includes the unique subject code and description, the number of units per subject (composed of lecture and laboratory units for computer subjects), status indicating that the subject is still offered, and special field where subjects marked as special such as research need only one (1) grade submission.

The *document table* hosts records of available documents that students can request. The records in this repository are pre-determined as only four (4) document types are available for request. It includes the document name and the cost per page of each document. In the document management module, the cashier can only update the prices of each document.

As an institution, the Academic Calendar of San Jose Community College is broken down into two Semesters and a Summer term. This academic year and term are recorded in the *schoolyear table*. For this table, the school year and academic term serve as its composite key. The system allows to have only one (1) active academic term which shall serve as the default academic term in navigating the sections, as well as the default academic year and term for section creation.

The personal and emergency contact information of personnel involved in the system will be stored in the system. Initially, the basic information of the personnel will be stored along with the account credentials given upon the creation of the account. The

access level of accounts varies depending on their role. The said access level serves as authorization for what functionalities will be available to the logged user. All accounts include a status field denoting whether an account can still be used to access the system.

For the student record, all necessary student personal information and the mother's full maiden name and contact will be stored in the system. As stated, the student's personal data and mother's maiden name will be used to authenticate the document request made by students online. The system also records the year of admission, year of graduation, NSTP, and special order serial number given to students. This information is retrieved and will be included in the generation of Transcript of Records.

The management of a section involves many data tables. Prior to the creation of the section, an entry to the subject offering table is recorded. The *subject offering table* comprises the academic term, subject, and instructor handling the subject. It is designed in this way because the same subject offering (handled by the same instructor) is offered to different student blocks. A section is created with the records retrieved from the subject offering, along with the data of the section name (composed of course, year level, and block). A default section criteria that houses the grade term weight is created.

The students' academic records are stored in the entity cluster of grades composed of *grade table*, *manual grade table*, *computed grade breakdown table*, and *grade removal table*. The base grade table contains the final grade and remarks incurred by the students. Normally, student grades undergo submission from instructors handling the subject and are duly approved by the registrar. These submissions are recorded in the *section grade submission table*. Submissions are made every grade term, and the particulars of the grades submitted every term for each student are recorded in the computed grade breakdown table. In addition, every term submission (resulting in the recording of student grades in the computed grade breakdown) updates the final grade in the grade table. Should the instructor or the registrar need to update the grade of students with Incomplete remarks, the removal grade will be stored in the grade removal table, associating the grade ID from the grade table.

In the case of the manual grade table, the entry of records in this table is done when

the registrar staff encodes the grades of students from the previous academic terms. Encoding student grades is done per section, although the system can cater to individual entry of grades, especially for recording the grades of transfer students.

The *notification table* is responsible for recording contents that will be used to provide reminders and updates to the respective users. For grade submissions, a click on the notification card will redirect the user to the corresponding section and term where the submission is made.

The request entity cluster is responsible for safekeeping the records of every scholastic request made in the system. It is composed of the *request*, *document request*, *document request term*, *document creation*, and *document history tables*. The base request table serves to store relevant data pertaining to the requestor, the date and validity of the request, the request status whether approved, rejected, or canceled, the status of payment, and the released field to identify if the requested documents are released. The document request table records what particular scholastic documents are requested in the request made. This includes the number of copies, the number of pages that a document has when generated, and its cost per page. In cases where a Certification of Grades is requested, the academic term is recorded in the document request term table.

Further, the purpose of the document creation table is to record what documents are generated and being processed. Vital information such as the date and time of generation, and who generated the document are recorded. The generated documents are then forwarded to the responsible units whose signatures are required for processing. The details of forwarding must also be recorded in the document history table. This is necessary in identifying the current whereabouts of the documents requested.

4.3 Construction

This is the stage in which the prototypes from the user design phase are integrated to form a working system. The conceptualized ideas during the previous phase and prototypes

were then transformed into actual system modules. Upon finalizing the appropriate data that the system needs through the use of data flow diagrams and Entity Relationship Diagram, the ERD is then converted into actual database tables that are well-defined and adhere to the principle of relational databases to facilitate efficient data storage and retrieval. Tables 4.1 to 4.25 serve as the reference guide for the construction of database tables.

Table 4.1: Data Dictionary: Department Table

Field Name	Data Type	Key	Field Size	Description	Example
dept_id	TINYINT	Primary Key	3	Unique Identifier Number for department	1
dept_name	VARCHAR		50	Name of the departments	College of Education
dept_code	VARCHAR		10	Short Code Department Identifier	CBE
head	INT	Foreign Key	10	Foreign Key Identifier	1

Table 4.2: Data Dictionary: Unit Table

Field Name	Data Type	Key	Field Size	Description	Example
unit_id	TINYINT	Primary Key	3	Unique Identifier Number for school units	1
unit_name	VARCHAR		50	Name of the school unit	Accounting Unit

Table 4.3: Data Dictionary: Course Table

Field Name	Data Type	Key	Field Size	Description	Example
course_id	TINYINT	Primary Key	4	Unique Identifier Number for courses	1
course_code	VARCHAR	Unique	10	Short-hand identifier for the programs offered	BSED
course_description	VARCHAR		100	Degree Programs offered at SJCC	BS in Secondary Education
major	VARCHAR		50	Specialization of Degree Program	Major in English
department_id	VARCHAR	Foreign Key	3	Foreign Key from Department Table	1

Table 4.4: Data Dictionary: School Table

Field Name	Data Type	Key	Field Size	Description	Example
school_id	TINYINT	Primary Key	3	Unique Identifier Number for schools	1
school_name	VARCHAR		255	Name of the school	San Jose Community College

Table 4.5: Data Dictionary: Subject Table

Field Name	Data Type	Key	Field Size	Description	Example
subject_code	VARCHAR	Primary Key	10	Unique Identifier Code for the subjects	FIL101
subject_description	VARCHAR		50	Descriptive title of the subject	Filipino 101
unit_lab	TINYINT		1	weight of the subject's laboratory unit	1
unit Lec	TINYINT		1	weight of the subject's lecture unit	3
status	TINYINT		1	numerical status: (1) active, (0) soft deleted	1
special	TINYINT		1	numerical status: (1) special subject like research, (0) non-special	1

Table 4.6: Data Dictionary: Document Table

Field Name	Data Type	Key	Field Size	Description	Example
document_id	TINYINT	Primary Key	3	Unique Identifier Number for the document	1
document_title	VARCHAR		50	Descriptive title of the document	Certificate of Grades
cost_per_page	DECIMAL		(6,2)	Cost of document per page	50.00

Table 4.7: Data Dictionary: Schoolyear Table

Field Name	Data Type	Key	Field Size	Description	Example
year	YEAR	Primary Key	4	Denoting the active school year	2021
term	VARCHAR	Primary Key	3	Denoting the active school term	1st
status	TINYINT		1	Numerical Status (1) when active school year and term	1
last_day	DATE			First day of the corresponding academic term	2021-06-15
last_day	DATE			Last day of the corresponding year and term	2021-12-15

Table 4.8: Data Dictionary: Staff Table

Field Name	Data Type	Key	Field Size	Description	Example
staff_id	INT	Primary Key	10	Identifier Number for the staff	1
staff_number	VARCHAR	Unique	10	Unique Number for the staff	2021-00001
first_name	VARCHAR		50	First name of the staff	Christian
middle_name	VARCHAR		50	Middle name of staff	Sanchez
last_name	VARCHAR		50	Last name of the staff	Lana
gen_suffix	VARCHAR		4	Generational suffix of the staff	Jr.
position	VARCHAR		50	Employment Position of the staff	Registrar
rank	VARCHAR		30	Academic Rank of the staff	Professor VI
department	TINYINT	Foreign Key	3	Foreign Key from Department table	1
unit	TINYINT	Foreign Key	3	Foreign Key from Unit table	1
bdate	DATE			Date of Birth of staff	1978-05-20
gender	VARCHAR		6	Gender of the staff	Female
civil_status	VARCHAR		9	Civil status of staff	Single
staff_cont_num	VARCHAR		10	Contact number of the staff	9951234567
staff_cont_person	VARCHAR		80	Name of emergency contact person of the staff	Richard M. Guevarra
staff_cont_per_num	VARCHAR		10	Contact number of the emergency contact person of staff	9307654321
relationship	VARCHAR		15	Relationship of the emergency contact person to the staff	Sibling

Table 4.9: Data Dictionary: Staff Education Table

Field Name	Data Type	Key	Field Size	Description	Example
staff_number	VARCHAR	Foreign Key	10	Foreign Key from Staff table	2021-00001
degree	VARCHAR		100	Degree program taken by the staff	BS in Secondary Education
type	VARCHAR		50	Degree level taken by the staff	College level
year_start	YEAR		4	Year started when taking the degree program	2015
year_end	YEAR		4	Year where staff graduated from the program	2019

Table 4.10: Data Dictionary: Account Table

Field Name	Data Type	Key	Field Size	Description	Example
id	INT	Primary Key	10	Unique Number assigned to user accounts	1
staff_number	VARCHAR		10	Unique identifier assigned to staff	2017-12345
username	VARCHAR	Unique	30	Unique identifier name for account authentication	us3rn@me.123
password	VARCHAR		255	User known password or phrase for account authentication	\$2y\$10\$RRwRGRF...
access_level	TINYINT		3	Number identifier for the staff's system privilege	1
status	TINYINT		3	Numerical status if staff is active or inactive	1
profile_pic	VARCHAR		255	Path for the staff's profile picture	\images\profilepic.png

Table 4.11: Data Dictionary: Student Table

Field Name	Data Type	Key	Field Size	Description	Example
student_id	INT	Primary Key	10	Identifier Number for the student	1
student_number	VARCHAR	Unique	10	Unique Number assigned to the student	2021-00001
first_name	VARCHAR		50	First name of the student	Christian
middle_name	VARCHAR		50	Middle name of student	Sanchez
last_name	VARCHAR		50	Last name of the student	Lana
gen_suffix	VARCHAR		4	Generational suffix of the student	Jr.
year_admission	YEAR		4	Year when a student is admitted	2015
year_graduated	YEAR		4	Year when a student has graduated	2015
remarks	VARCHAR		30	Status of the student	Transferee
bdate	DATE			Date of Birth of student	1995-05-20
so_date	DATE			Date of conferment	1995-05-20
so_number	VARCHAR		30	CHED Special Order Number	2021-00001
nstp_serial	VARCHAR		30	CHED Issued NSTP Serial	2021-00001
gender	VARCHAR		6	Student's gender	Female
civil_status	VARCHAR		9	Civil status of student	Single
stud.cont.num	VARCHAR		10	Contact number of the student	9951234567
mother_maiden_name	VARCHAR		100	Full name of student's mother's maiden name	Richard M. Guevarra

Table 4.12: Data Dictionary: Subject Offering Table

Field Name	Data Type	Key	Field Size	Description	Example
offering_id	INT	Primary Key	10	Unique identifier number assigned to subjects offered	1
schoolyear	YEAR	Foreign Key	4	Foreign key from the schoolyear table	2015
term	VARCHAR	Foreign Key	3	Foreign key from the schoolyear	1st
subject_code	VARCHAR	Foreign Key	10	Foreign Key from the Subject table	FIL101
staff_id	INT	Foreign Key	10	Foreign key from the Staff table	1

Table 4.13: Data Dictionary: Section Table

Field Name	Data Type	Key	Field Size	Description	Example
section_id	INT	Primary Key	10	Unique identifier number assigned to sections	1
section_code	VARCHAR	Unique	20	Unique identifier code for each section	2018-12-BSED1A
offering_id	INT	Foreign Key	10	Foreign Key from the Subject Offering table	1
schoolyear	YEAR	Foreign Key	4	Foreign key from the schoolyear table	2015
term	VARCHAR	Foreign Key	3	Foreign key from the schoolyear	1st
section_name	VARCHAR		20	Block section name	BSED-1A
grade_term	TINYINT		1	Number of term for the subject offered	4
man_input	TINYINT		1	Indicator if the grades in the section are encoded by the registrar staff	1

Table 4.14: Data Dictionary: Section Criteria Table

Field Name	Data Type	Key	Field Size	Description	Example
section_id	INT	Foreign Key	10	Foreign Key from the Section table	1
Prelim	TINYINT		4	Weight of the Prelim period	25
Midterm	TINYINT		4	Weight of the Midterm period	25
Prefinal	TINYINT		4	Weight of the Prefinal period	25
Final	TINYINT		4	Weight of the Final period	25

Table 4.15: Data Dictionary: Section Grade Submission Table

Field Name	Data Type	Key	Field Size	Description	Example
section_id	INT	Foreign Key	10	Foreign Key from the Section table	1
term	TINYINT		3	indicates the term period when the grades are submitted	1
status	TINYINT		3	number identifier if grades are submitted, approved, or rejected	1

Table 4.16: Data Dictionary: Grade Table

Field Name	Data Type	Key	Field Size	Description	Example
grade_id	BIGINT	Primary Key	20	Unique identifier for the student grades	1
student_id	BIGINT	Foreign Key	20	Foreign Key from the student table	1
final_grade	DECIMAL		(5,2)	indicates the performance of student with respect to the subject	85.00
remarks	VARCHAR		20	Indicates student's remark whether passed, failed, dropped, etc.	Passed
grade_entry	TINYINT		3	numeric identifier whether the final grade is manually inputted or by system processes	1

Table 4.17: Data Dictionary: Manual Grade Table

Field Name	Data Type	Key	Field Size	Description	Example
grade_id	BIGINT	Foreign Key	20	Foreign key from grade table	1
subject_code	VARCHAR	Foreign Key	20	Foreign Key from the subject table	1
schoolyear	YEAR		4	records the year when the subject is taken	2020
term	VARCHAR		3	records the term when the subject is taken	1st
school_id	INT	Foreign Key	10	Foreign key from the School table	1
out_subjCode	VARCHAR		10	Code used for subjects taken outside of SJCC	BUFIL101
out_subjDesc	VARCHAR		50	Descriptive title of the subject taken outside of SJCC	Komunikasyon sa Akademikong Filipino

Table 4.18: Data Dictionary: Computed Grade Breakdown Table

Field Name	Data Type	Key	Field Size	Description	Example
grade_id	BIGINT	Foreign Key	20	Foreign key from grade table	1
section_id	INT	Foreign Key	10	Foreign Key from the section table	1
grade	TINYINT		3	student grade garnered per term	85
term	TINYINT		3	denotes on what term period is the grade record	1
weight	TINYINT		3	denotes the weight of the particular term period	25
remarks	VARCHAR		20	Indicates whether a student had passed, failed, etc. on the term period	1

Table 4.19: Data Dictionary: Grade Removal Table

Field Name	Data Type	Key	Field Size	Description	Example
grade_id	BIGINT	Foreign Key	20	Foreign key from the grade table	1
removal	TINYINT		3	Grade given to conditionally passed students after submitting the requirements	80

Table 4.20: Data Dictionary: Notification Table

Field Name	Data Type	Key	Field Size	Description	Example
notif_id	INT	Primary Key	10	Unique identifier for the notifications	1
staff_id	INT	Foreign Key	10	Foreign key from the staff table	1
notif_link	VARCHAR		255	Contains the link for action	grades?section_code=2018-13-BSCS1A#Prelim
content	VARCHAR		255	Contains the information regarding grade and document request	Submission for 2018-13-BSCS1A Prelim period has been made
notif_date	DATE			Date when a request or grade related matter is made	2020-01-12
notif_type	TINYINT		3	numerical indicator is notification is grade- or request-related	1
notif_read	TINYINT		3	numerical indicator whether the notification is read or not	1

Table 4.21: Data Dictionary: Request Table

Field Name	Data Type	Key	Field Size	Description	Example
request_id	INT	Primary Key	10	Unique Identifier Number for the request	1
student_number	VARCHAR	Unique	10	Unique Number assigned to the student	2021-00001
grad_year	YEAR		4	Year when a student has graduated	2015
contact	VARCHAR		10	mobile number of the requester	9901234567
request_date	DATE			Date when a request was made	2021-05-20
validity_date	DATE			Up to when will the request be valid	2021-05-20
verification_code	VARCHAR		8	A code given that will be used for tracking the request	ABCDE132
status	TINYINT		1	Numerical indicator of request whether pending, approved, rejected, or cancelled	1
paid	TINYINT		1	Numerical indicator of request whether paid or pending payment	1
released	TINYINT		1	Numerical indicator of request whether released or unreleased	1

Table 4.22: Data Dictionary: Document Request Table

Field Name	Data Type	Key	Field Size	Description	Example
request_id	INT	Primary Key	10	Unique identifier for the requests	1
document_id	TINYINT	Foreign Key	3	Foreign key from the document table	1
num_copy	TINYINT		3	Number of copies requested per document	1
num_pages	TINYINT		3	Number of pages the requested document will have	1
cost_pages_page	DECIMAL		(6,2)	Cost of each page from the time the document is requested	50.00

Table 4.23: Data Dictionary: Document Request Term Table

Field Name	Data Type	Key	Field Size	Description	Example
request_id	INT	Foreign Key	10	Foreign key from the request table	1
document_id	TINYINT	Foreign Key	3	Foreign key from the document table	1
req_year	YEAR			Indicates the year where grade records will be fetched	2021
term	VARCHAR		3	Indicates the term through which the grade records will be fetched	1st

Table 4.24: Data Dictionary: Document Creation Table

Field Name	Data Type	Key	Field Size	Description	Example
doc_created_id	INT	Primary Key	10	Unique identifier for the document created	1
request_id	INT	Foreign Key	10	Foreign key from the request table	1
document_id	TINYINT	Foreign Key	3	Foreign key from the document table	1
doc_creation_datetime	DATETIME			Records the date-time when a document was created	2020-03-26 14:31:39
remarks	TINYINT		3	Numerical indicator whether a document is in circulation, done, or cancelled	1
staff_id	INT	Foreign Key	10	Foreign key from the staff table	1

Table 4.25: Data Dictionary: Document History Table

Field Name	Data Type	Key	Field Size	Description	Example
doc_created_id	INT	Foreign Key	10	Foreign Key from the Document Creation table	1
receiver_id	INT	Foreign Key	10	Foreign key from the staff table	1
forwarded_id	INT	Foreign Key	10	Foreign key from the staff table	1
datetime_forWARDED	DATETIME			Records the date-time when a document was forwarded	2020-03-26 14:31:39

The system was developed incrementally, starting from the most crucial set of modules. Several tests were conducted with the registrar to ensure that everything worked and that the result satisfied the client's expectations and system objectives.

4.4 Cutover

This is the final stage for deploying the system on a selected online database server. The phase also includes user orientation and training.

4.4.1 Implementation Plan

The researcher will be using the Pilot implementation method. In this method, the developed system will be implemented simultaneously with the existing system [19]. Data from the existing system will be migrated to the new system. When the management has adapted to its use, and ensured that the system is functioning properly according to their needs, the old system is discontinued.

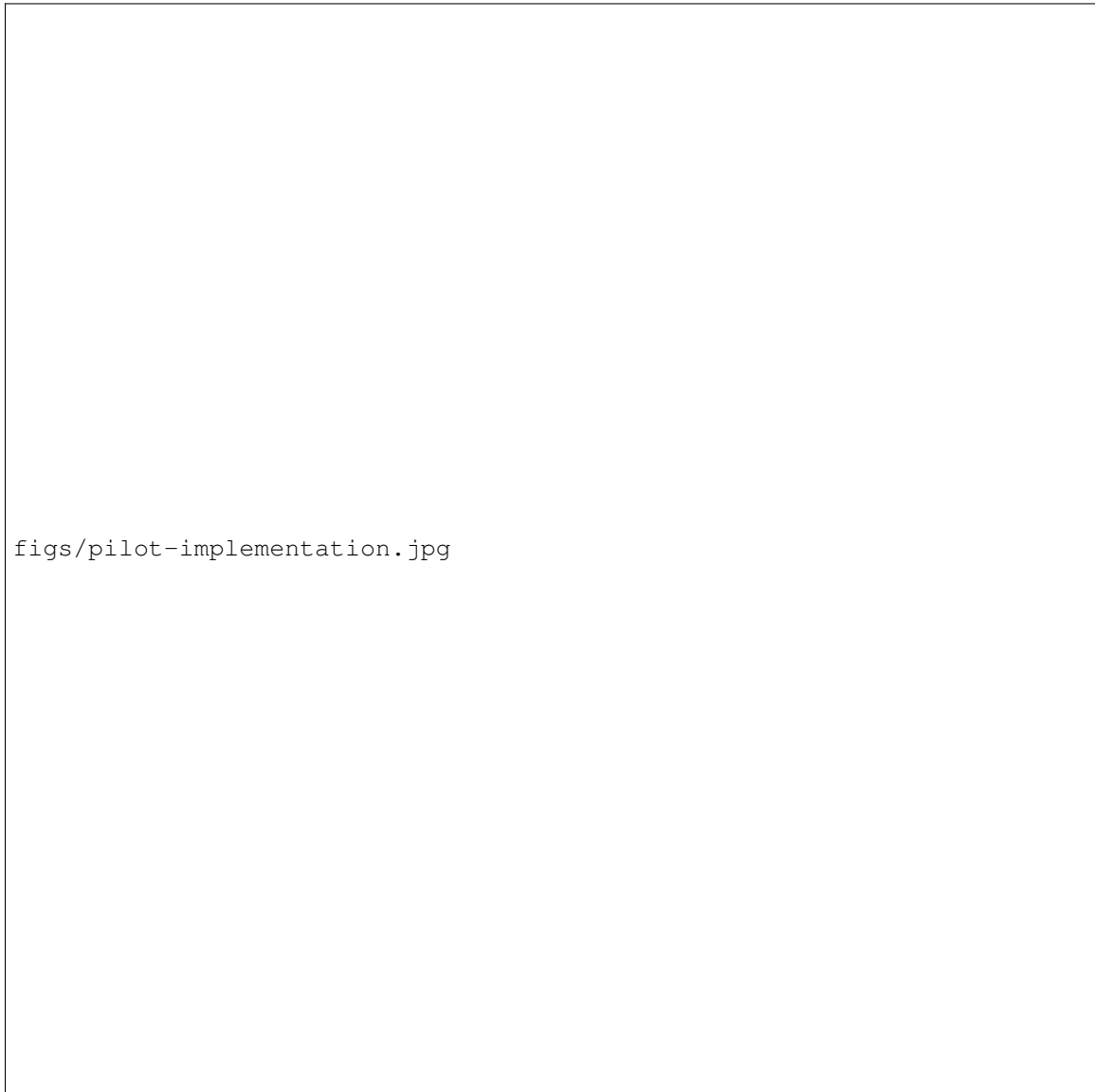


Figure 4.23: Pilot Implementation

The graphic above depicts a progressive shift from the old to the new system. The new system will be implemented and made available to the intended users. The goal is for users to learn about and assess the new system.

4.4.2 User's Training Plan

One of the most crucial elements for a successful system implementation is training the end users. Before delivering a hands-on training session to all users across the university, the researcher intends to train the IT staff, registrar, and faculty president. Table 4.26 shows the particulars of the training plan.

Table 4.26: Users Training Plan

Department	Schedule
IT Staff, Registrar, Faculty President	December 18-20, 2023
Administration Office	December 18-20, 2023
Registrar's Office	December 18-20, 2023
College of Education	December 18-20, 2023
College of Business Education	December 18-20, 2023
IT Unit	December 18-20, 2023

4.4.3 Maintenance Plan

Once the system is implemented, the researcher utilizes corrective and preventive maintenance.

Figure 4.24 shows a template of a bug or incident report that will be given to the institution. Users who may encounter bugs relating to the system can use the template and file the same to the IT unit. The IT unit will then be responsible for communicating with the researcher to resolve the bug. The patched source code will then be submitted to the IT unit responsible for updating the source code in the server.

For preventive maintenance, the system included a module where the IT Unit is responsible for providing a backup of the entire database. This backup shall be utilized in cases where problems in the database arise.

Bug Incident Report	
1. Report ID:	
2. Reported by:	
3. Department:	
4. Location:	
5. Position:	
6. Date:	
7. Please describe the steps to reproduce the problem	
8. Please describe the observed result from the system	
9. Please describe the expected/correct result.	
10. Screen capture <i>For bug report, please include a screen capture of the error</i> 1. Navigate to the page where the error is experienced 2. Press the printscreen or use a snipping tool 3. Paste the screen capture along with this Bug Incident Report	

Figure 4.24: Bug or Incident form

4.5 Development Log

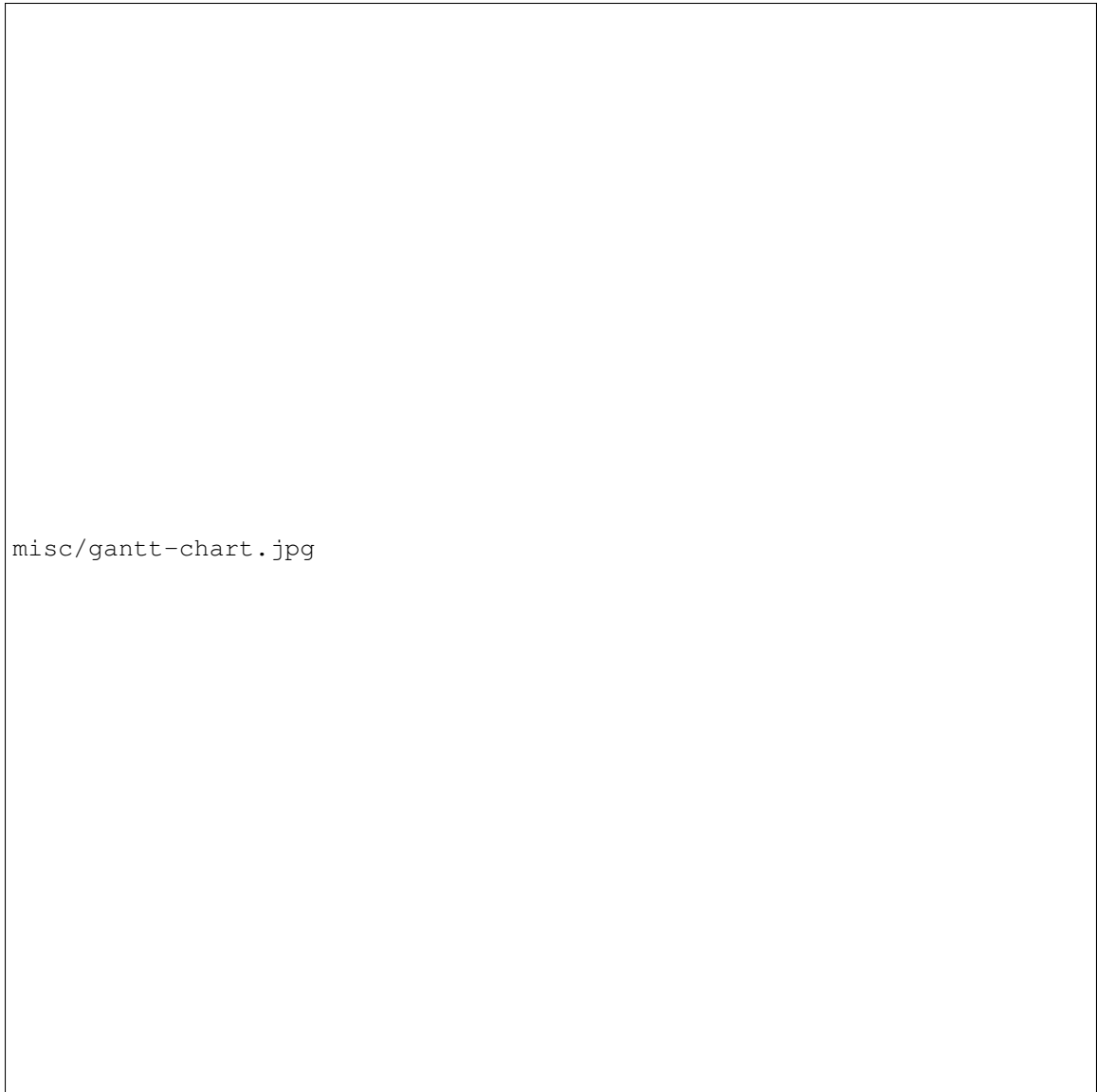


Figure 4.25: Development Log Frame

The proponent took an estimated eighth (8) months to develop and turn over the system. The proponent started with the necessary information needed prior to the start of the development. This includes interviews with personnel, both teaching and non-teaching, as well as students. The initial phase also covered a discussion of issues that need resolution through the use of a system. The proponent carefully reviewed the deliberations of the interviews to properly plan the functionalities relevant to the system.

The user design phase is responsible for presenting prototypes through wireframes and initial webpage interfaces that will be used as a basis for the development of the system. The prototypes are assessed by the end users for feedback to be encouraged and understood for their successful implementation.

Once prototypes and feedback were collated, the proponent progressed to the construction phase. This phase is expected to take the most development time as it is responsible for the actual development of the system functionalities. In this phase, the proponent is focused on completing the major functionalities and components that the client needs for the operations. The approved modules are incrementally added to the system. From time to time, the system is presented to the end-user for evaluation.

The last phase of the methodology is allotted for the preparation of user manuals, training, and implementation of the system. The provision of user manuals and training of users is targeted for December, while the hosting of the system is expected in January 2024 due to its need for funding.

4.6 Resources

The following resources helped keep this project going and reach the objectives of the study.

4.6.1 Respondents

The respondents of the study consisted of one selected individual from each role within the company: the owner, a production staff, and a warehouseman. They were the respondents that the researchers had approached because they were the ones who had sufficient knowledge regarding the process of the company. In order to gather the needed information from the respondents, the proponents had prepared an interview questionnaire, which the respondents answered according to what they know about what they asked. In addition, the proponents asked for documents from the respondents that were required for use in the development of the system. To know more about the role of the respondent mentioned above and to better understand why they were chosen as respondents, the proponents listed their information on what kind of role the respondents should maintain.

- **Owners**

The Owner serves as the head of the company, responsible for overseeing and managing various aspects of the business. This includes tasks such as checking the inventory storage to ensure optimal stock levels and efficient management, as well as monitoring the status of the products.

- **Production Staffs**

The responsibility of the Production Staff will manage the manufacturing processes to ensure that the products meet the specified requirements and standards.

- **Warehouseman**

The Warehouseman's responsibility included checking and recording the items stored in the inventory, This meant carefully checking and listing every item to make sure everything will be in order.

Table 4.27: The Respondents

Respondents	Frequency
Owner	1
Production staff	1
Warehouse Personnel	1
Total Respondents	3

4.6.2 Technical Resources

The main purpose of technical resources will be to modify the necessary hardware and software to develop the proposed system's specifications. For the development of the proposed system, the proponents evaluated the following essential resources:

4.6.3 Hardware Resources

a. Development Resources

Hardware	Minimum Requirements
Laptop	AMD A-46300 APU Radeon (tm) 4 GB of RAM 260 GB of storage 32 bit Operating System
Internet Connection	at least 10 mbps

b. Deployment Resources

Hardware	Minimum Requirements
Desktop Computer or laptop	Intel(R) Pentium(R) CPU B940 CPU up to 1.80GHz 4.00 GB of RAM
Android	Android 10 4.00 GB of RAM
Internet	at least 10 mbps
Tablet	Android 11, ZUI 12.5

4.6.4 Software Resources

a. Development Resources

Software	Recommended Requirements
Windows10	Browser Chrome, Firefox, Edge, IE11+, Safari 9.1, Opera Recommended Browser: Chrome Xampp 8.2 or later

Software	Recommended Requirements
Windows10	Browser Chrome, Firefox, Edge, IE11+, Safari 9.1, Opera Recommended Browser: Chrome Xampp 8.2 or later
Android	Android 11 (XOS 7.6)
Tablet	Android 11, ZUI 12.5

4.7 Research

4.7.1 Instruments

The Primary and Secondary data were collected for this study. The primary data will be collected through interviews, questionnaires, and acquired forms. The secondary data will be gathered from the Internet, online papers such as journals, and websites related with the proposed system.

Interview Questionnaire

The researchers employed an interview questionnaire as a means of collecting data, aiming to minimize the time respondents needed to spend responding to written questions in the questionnaire. For this method of data collection, the researchers assist the respondents by reading the questions and then writing down or recording their responses with their prior consent. Also, the researcher conducted follow-up interviews to gather additional information.

4.7.2 Statistical Tools

Statistics mathematical techniques encompass gathering, analyzing, and organizing data. Total enumeration sampling, a form of purposive sampling, involves the researcher examining the entire population that has specific characteristics. In the context of sampling, units refer to the elements constituting the population. In total enumeration sampling, it is highly probable that these units will be individuals [?].

The system's evaluation involved three respondents: the owner, the warehouseman, and a production staff. They filled out survey questionnaires with responses ranging from "strongly agree" to "strongly disagree," scored from 4 to 1, respectively. The range of answers is designed in this manner for the respondent's sentiment is essential without being neutral. The researcher then calculated a weighted mean to measure the general response of the different users. The researcher used a weighted mean to measure the general response of the different users.

$$WeightedMean = \frac{F1(4) + F2(3) + F3(2) + F4(1)}{F1 + F2 + F3 + F4}$$

Where:

F1 = number of respondents who strongly agree

F2 = number of respondents who agree

F3 = number of respondents who disagree

F4 = number of respondents who strongly disagree

In terms of interpreting the result, the researcher used the uniform interval of the Four (4)-point Likert scale 5.1. The uniform interval in a sense eliminates the possible bias that may produced during the survey. Table 5.1 shows the of having the majority uniform difference in each interval except for one.

Likert Scale	Interval	Description
4	xxx - xxx	Strongly Agree
3	xxx - xxx	Agree
2	xxx - xxx	Disagree
1	xxx - xxx	Strongly Disagree

Table 4.28: Four-point Likert Scale

4.7.3 Cost and Benefit Analysis

This section shows a difference in the cost of an alternative system compared to the proposed system. In this case, the advantages of alternative systems, proposed systems and current systems are shown by means of an approximation and total of all equivalent monetary values. The price of the alternative system that will be used in the tables for cost-benefit analysis will be provided by BoomText Team from Boom AI.

It shows the comparison between the alternative system, proposed system, and current system reveals the differences in expenses. This analysis outlines the advantages of each system by estimating and totaling their equivalent monetary values. The cost of the alternative system, used in the tables for the analysis, is provided by BoomText Team from Boom AI.

Table 4.29: Proposed System VS. Alternative System Development Cost

	Proposed System	Alternative Cost
Development	Free	P 190,000

For the development of the proposed system, the proponents develop the system for free while for the alternative system, the fee summary for the development of the alternative system cost P190,000.

Table 4.30: Manpower Daily Rates

	Proposed System	Alternative Cost
Developer 1 - Backend	FREE	35,000/month
Developer 2 - Frontend		35,000/month
Project Manager		25,000/month
System Analyst		20,000/month
UI/UX		20,000/month
Total Manpower Monthly Fee:		P 135,000

Concerning the monthly fee for system development, the system proponents have chosen not to include any charges throughout the development phase. Consequently, they took on the roles of both frontend and backend developers to build the envisioned system. Alternatively, the monthly fee for each developer will be established based on their specialization and the specific development project that the researchers are working on. The earlier mentioned monthly rates were calculated using the initial cost for each field within the company, and any extra fees for the system were also determined from these set monthly rates.

Table 4.31: Other Charges for The Deployment of the System

Task	Proposed System	Alternative Cost
Maintenance	FREE	25,000/month
Server Set Up Fee		25,000/month
Documentation		5,000/month
Total		P 55,000

The total expense for the monthly of additional services rendered during the system development will be calculated based on the monthly rates for manpower. This

calculation enabled us to determine the overall cost of the monthly rate for supplementary services offered in the alternative system development. As for the planned system, the creators assured that they furnish all essential documentation and services without any associated fees.

Table 4.32: Proposed System VS Alternative System Annual Operational Cost

Current System		Proposed System		Alternative System	
Unit	Amount	Unit	Amount	Unit	Amount
Valiant Record Book (5 Record book)	P 590	Maintenance Fee	P 3,000	Alternative (package)	P 190,000
Ink Dye For Epson L210	P 10,920	Ink Dye For Epson L210	P 10,920	Ink Dye For Epson L210	P 10,920
Hard Copy Bond Paper (20 rims) Long	P 4,160	Hard Copy Bond Paper (20 rims) Long	P 4,160	Hard Copy Bond Paper (20 rims) Long	P 1,699
Hard Copy Bond Paper (20 rims) Short	P 3,560	Hard Copy Bond Paper (20 rims) Short	P 3,560	Hard Copy Bond Paper (20 rims) Short	P 3,560
Calculator 12 Digits	P 449				
Panda Ballpen (1box)	P 120	Pentel Pen HBW(1box)	P 99	Pentel Pen HBW(1box)	P 99
Pentel Pen HBW(1box)	P 99	Hostinger (Business Plan)	P 5,628	Hostinger (Business Plan)	P 5,628
Pencil Mong-gol 1 (1box)	P 119	Internet at least 10mbps	P 15,588	Internet at least 10mbps	P 15,588
Lenovo LOQ 15IRH8	P 56,635				
Internet at least 10 mbps	P 15,588	Domain Name (.com)	P 679	Domain Name (.com)	P 679
Total	P 92,320		P 43,634		P233,634

This table shows the yearly operational expenses and costs associated with Weavemanila's existing system, the alternative system, and the proposed system if adopted or put into practice. In gathering the prices for the mentioned items, the researchers used online stores to ascertain the costs of each item, including pencils, Pentel pens, printer ink, bond paper, calculator, record books. Proponents selected the hosting services based on prices listed on the Hostinger website. These prices were obtained through research conducted on various online computer shops and school supply stores on Shopee. The price of the alternative system that will be used in the tables for cost-benefit analysis is provided by BoomText Team of Boom AI.

Referring to the table above, the current system cost us 92,320 pesos, while the proposed system comes in at 34,314 pesos, and the Alternative system is a whopping 224,314 pesos. Even though the proposed system is pricier than what we currently use, it brings a lot of advantages, making work easier, saving time, and boosting the efficiency of Weavemanila's overall services. The researchers figured out the difference in the annual operational cost using the calculator on our local computers, following the formula provided below.

CS-Current system
 PS-Proposed system
 AS-Alternative System
 cs=92,320
 ps=34,314
 as=224,314

annual operation cost difference for current system and proposed

Formula : $p = \frac{cs - ps}{cs}$
 $cs - ps = 92,320 - 34,314 / 92,320$
 $= 58,006 / 92,320 = 0.6283$
Increase = 62.83%

Chapter 5

Contributions and Recommendations

5.1 Summary of Contributions

This chapter discusses how the Online Production Monitoring and Inventory Management System could help WeaveManila. Currently, WeaveManila faces challenges, especially with keeping track of materials and organizing data. They use separate Excel files for different things, which makes it hard to manage everything.

With the new system, WeaveManila can track purchases and store all information in one place. This will make it easier for them to manage their inventory and production processes. By automating manual tasks and ensuring data accuracy, the system will help WeaveManila work more efficiently and make fewer mistakes. Overall, implementing this system could greatly improve WeaveManila's operations and decision-making.

5.2 Client Test Result, Analysis and Interpretation

Prototyping and iterative design processes with continuous user and target audience feedback are made possible by testing while the product is being developed. The end-users, and developer communicated consistently throughout the process of the development [24].

5.3 Results and Discussion

This section shows the result of the overall test. Tables from 5.2 to 5.4 demonstrate the collated survey results from the different users grouped by functionality, speed, design, and usability.

Functionality	4	3	2	1	W. Mean
Q1	3	0	0	0	4
Q2	3	0	0	0	4
Q3	3	0	0	0	4
Q4	2	1	0	0	3.67
Q5	0	3	0	0	3
Q6	0	3	0	0	3
Q7	0	3	0	0	3
Q8	1	2	0	0	3.33
Average	3.5				

Table 5.1: Overall Functionality Summary

The table 5.1 shows the overall results of 3 respondents. In the functionality category, questions Q1 to Q3, have the equal weighted mean result which is Strongly Agree. The respondents found the tested system easy to use in terms of logging in to the system and using (MPO) to the system.

Speed	4	3	2	1	W. Mean
Q1	2	1	0	0	3.67
Q2	1	2	0	0	3.33
Q3	0	3	0	0	3
Q4	1	2	0	0	3.33
Average	3.33				

Table 5.2: Overall Speed Summary

The table above shows results for the speed category. Among these, the question with the highest weighted mean score has Agree remark. This result that users found the system minimizing the time it takes compared to the current system.

Usability	4	3	2	1	W. Mean
Q1	1	2	0	0	3.33
Q2	3	0	0	0	4
Q3	0	3	0	0	3
Q4	0	3	0	0	3
Average	3.33				

Table 5.3: Overall Usability Summary

The usability results in the table above indicate that the system's chosen color combinations and fonts have received an agreeable score, with respondents concurring that they are suitable for the system's purpose. This may entail necessary measures to improve the chosen color and fonts.

	Owner	Warehouseman	Production staff	W. Mean
Functionality	3.63	3.25	3.5	3.5
Speed	3.25	3.5	3.25	3.33
Usability	3.25	3.25	3.5	3.33
Overall Average	3.29	3.46	3.42	3.39

Table 5.4: Overall Summary

Table 5.5 illustrates the general consensus of the owner, warehouseman, and production staff. In this table, the average functionality score for all users resulted in 3.39, suggesting that the general users strongly agree that the intended functionality of the system is met. The test results attest that the targeted purpose of developing the system is achieved.

5.4 Conclusions

The researchers concluded that:

- The warehouseman can easily create purchase order forms and send them to the supplier's email, as well as view the status of purchased materials and inventory stocks.
- The production staff can easily view estimated completion dates for each project, including the current status of their job orders.
- The system is capable of storing and retrieving data, as well as implementing a monthly backup system for Weavemanila data to ensure recovery in case of system corruption.

5.5 Recommendations

The following are the recommendations of the client and developer to further enhance the system:

- Enhance the design interface further to make it easily understandable for the user.
- Add summary reports for weavers to see who has accomplished the most within a year.

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